READING FIRE DEPARTMENT, ENGINE 7

950 MCKNIGHT ST, READING, PA 19601

HVAC REPLACEMENT / INDOOR AIR QUALITY UPGRADE

BID DOCUMENTS: 22 JULY 2022

GENERAL NOTES:

1. ALL CONTRACTORS SHALL BECOME FAMILIAR WITH THE EXISTING CONDITIONS AFFECTING THIS PROJECT AND COORDINATE WITH OTHER DISCIPLINES.
2. THE WORK IN ACCORDANCE WITH DRAWINGS AND SPECIFICATIONS SHALL CONSIST OF PROVIDING ALL EQUIPMENT, MATERIALS, LABOR, SERVICES AND PERFORMED IN COMPLIANCE WITH THE APPLICABLE CODES AND STANDARDS.
3. ALL EQUIPMENT SHALL BE HANDLED, STORED, AND PROTECTED TO PREVENT DAMAGE BEFORE AND DURING INSTALLATION IN ACCORDANCE WITH THE MANUFACTURER'S INSTRUCTIONS.
4. ALL EQUIPMENT SHALL BE INSTALLED AND ADEQUATE CLEARANCES BE PROVIDED IN ACCORDANCE WITH THE MANUFACTURER'S INSTRUCTIONS AND CODE.
5. ALL NEW EQUIPMENT SHALL BE TESTED IN ACCORDANCE WITH APPLICABLE CODES AND STANDARDS. LEAKS, IF ANY, SHALL BE REPAIRED AND THE PIPING SHALL BE RETESTED TO MEET THE REQUIREMENTS.
6. ALL NEW EQUIPMENT IS SHOWN IN APPROXIMATE POSITION. THE CONTRACTOR SHALL FIELD VERIFY THE ROUTING AND TIE-INS OF NEW AND EXISTING PIPING. ALL NEW PIPING SHALL BE INSTALLED AND ADEQUATELY SUPPORTED IN ACCORDANCE WITH APPLICABLE CODES AND STANDARDS.
7. ALL CONTRACTORS SHALL CHECK AND FIELD VERIFY ALL CONDITIONS AND DIMENSIONS AT THE SITE PRIOR TO SUBMITTING BID AND BEFORE START OF CONSTRUCTION.
8. THE MECHANICAL CONTRACTOR IS RESPONSIBLE FOR COORDINATING INSTALLATION OF ALL EQUIPMENT SHOWN ON THE DRAWINGS, INCLUDING COORDINATION IF ANY EQUIPMENT OF ALTERNATE MANUFACTURER. THE CONTRACTOR SHALL PROVIDE COMPOSITE DRAWINGS AS REQUIRED FOR THE INSTALLATION OF EQUIPMENT AS SHOWN ON PLAN FOR APPROVAL BY ENGINEER.
9. ANY EQUIPMENT, MATERIALS, LABOR OR SERVICES NOT SPECIFICALLY MENTIONED HEREIN WHICH MAY BE NECESSARY TO COMPLETE OR PERFECT ANY PART OF INSTALLATION IN A SUBSTANTIAL MANNER SHALL BE FURNISHED WITHOUT EXTRA COST TO THE OWNER.
10. ALL WORK SHALL COMPLY WITH LOCAL AND NATIONAL CODES AND STANDARDS, UNDERWRITERS LABORATORY APPROVAL, AND ALL STATE AND FEDERAL OSHA SAFETY REQUIREMENTS.
11. LOCAL CODES SHALL SUPERCEDE INTERNATIONAL BUILDING CODES WHERE CONFLICTS OCCUR.
12. PROVIDE DIELECTRIC UNIONS AND PROPERLY ISOLATE CONTACT BETWEEN DISSIMILAR METALS TO INHIBIT GALVANIC CORROSION.
13. THE CONTRACTOR SHALL OBTAIN AND PROVIDE OWNER'S REPRESENTATIVE WITH NECESSARY PERMITS PRIOR TO CONSTRUCTION.
14. LABEL AND/OR PAINT ALL NEW EQUIPMENT, PIPING, INSTRUMENTS, WIRING, CONDUIT, AND ACCESSORIES.
15. PROVIDE WELD-O-LETS FOR WELDED PIPING. "FISH-MOUTH" CONNECTIONS ARE NOT ACCEPTABLE.
16. IF SHUT-OFF OF ANY UTILITY IS REQUIRED, COORDINATE ALL EFFORTS WITH THE RESPECTIVE UTILITY PROVIDER:
    MET ED FIRST ENERGY GROUP: KEN SWEITZER AT 610-921-6568
    UGI UTILITIES, INC.: KURT ZIELASKOWSKI AT 610-736-5571
    READING AREA WATER AUTHORITY: ANTHONY STEINLE AT 610-406-6300

APPLICABLE CODES:

1. PENNSYLVANIA UNIFORM CONSTRUCTION CODE (2018 ICC)
   - INTERNATIONAL EXISTING BUILDING CODE 2018 (IEBC 2018)
   - INTERNATIONAL FIRE CODE 2018 (IFC 2018)
   - INTERNATIONAL PLUMBING CODE (IPC 2018)
   - INTERNATIONAL MECHANICAL CODE (IMC 2018)
   - INTERNATIONAL FUEL GAS CODE (IFGC 2018)
   - INTERNATIONAL ENERGY CONSERVATION CODE (IECC 2018)
   - INTERNATIONAL WILDLAND-URBAN INTERFACE CODE (IWUIC 2018)
   - INTERNATIONAL PERFORMANCE CODE (ICCPC 2018)
   - INTERNATIONAL ACCESSIBLE AND USEABLE BUILDING AND FACILITIES CODE (a117.1,2017)
   - NATIONAL ELECTRICAL CODE 2017 (NFPA 70, 2017)
   - BUILDING CODE REQUIREMENTS FOR STRUCTURAL CONCRETE (ACI 318 2014)

DEMOLITION NOTES:

1. THE CONTRACTORS SHALL OBTAIN ALL REQUIRED PERMITS AND TURN OFF ALL IMPACTED UTILITIES BEFORE STARTING WORK.
2. ALL DEMOLITION/REMOVAL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE STANDARDS AND REGULATIONS.
3. DEMOLITION WORK SHALL BE SCHEDULED AND IMPLEMENTED WITH MINIMAL DISRUPTION TO ADJACENT OCCUPIED AREAS.
4. REMOVE ALL MATERIALS IN A SAFE WORKMANLIKE MANNER AND DISPOSE OF PER ALL APPLICABLE CODES & SAFETY REQUIREMENTS.
5. CONSTRUCTION DEBRIS MUST BE REMOVED FROM OCCUPIED SPACES AT THE END OF EACH WORK SHIFT.

OWNER

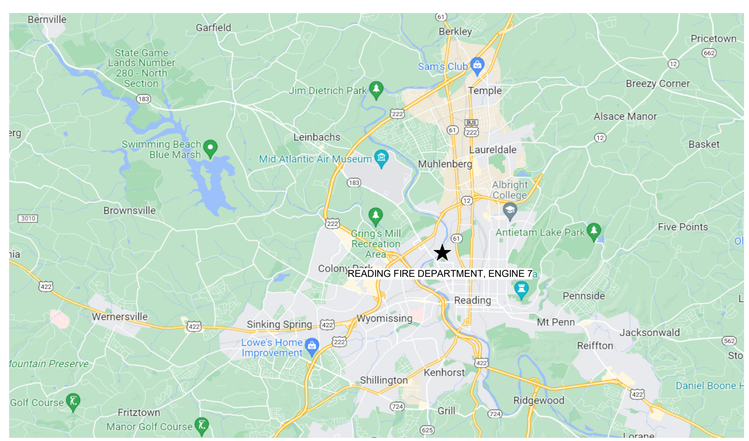
CITY OF READING
899 MORGANTOWN ROAD
READING, PA 19607
PHONE: 610-655-6502
EMAIL: DAVID.ANSPACH@READINGPA.GOV
ATTN: DAVID ANSPACH, PROJECT ADMINISTRATOR

ENGINEER OF RECORD

GANNETT FLEMING, INC.
1010 ADAMS AVENUE
VALLEY FORGE, PA 19403
PHONE: 610.650.8101
FAX: 610.650.8190
EMAIL: BWEISSER@GFNET.COM
ATTN: BRIAN M. WEISSER, P.E.

BUILDING DATA:

ORIGINAL CONSTRUCTION:
2 STORIES
CONSTRUCTION TYPE:
TYPE II-B:
UNPROTECTED-NONCOMBUSTIBE



LOCATION MAP NOT TO SCALE

DRAWING LIST

SHEET #	DRAWING #	SHEET NAME
1	G001	PROJECT COVER SHEET
2	M001	MECHANICAL COVER SHEET
3	M002	GENERAL NOTES AND ABBREVIATIONS
4	M101	DEMOLITION BASEMENT AND FIRST FLOOR - HVAC
5	M102	NEW WORK BASEMENT AND FIRST FLOOR - HVAC
6	M103	MECHANICAL DEMOLITON ROOF PLAN
7	M104	MECHANICAL NEW WORK ROOF PLAN
8	M501	DETAILS I
9	M502	DETAILS II
10	M601	SCHEDULES
11	M801	MECHANICAL CONTROL DIAGRAMS I
12	M802	MECHANICAL CONTROL DIAGRAMS II
13	M803	MECHANICAL CONTROL DIAGRAMS III
14	E001	LEGENDS AND ABBREVIATIONS
15	E101	DEMOLITION BASEMENT AND FIRST FLOOR
16	E102	NEW WORK BASEMENT AND FIRST FLOOR
17	E103	ROOF PLAN

GANNETT FLEMING

Gannett Fleming, Inc.
1010 Adams Avenue
Valley Forge • PA • 19403
t 610 • 650 • 8156
www.gannettfleming.com

READING FIRE DEPARTMENT, ENGINE 7
950 MCKNIGHT ST, READING, PA 19601
HVAC REPLACEMENT / IAQ UPGRADE

NO.	DATE	DESCRIPTION

REVISIONS

BID DOCUMENTS

PROJECT NUMBER: 066446.007

DESIGNED:	MODELED:	CHECKED:
SAJ	AMF	MJM

DATE: 22 JULY 2022

DRAWING TITLE: PROJECT COVER SHEET

DRAWING NO.: G001 SHEET 1 OF 17

Local File: BIM 360://066446-City_of_Reading/066446_007_MEP_Central_R21.rvt
Project Number: 066446.007
7/22/2022 6:59:37 PM

LINES & REFERENCE SYMBOLS

- NEW WORK LINEWEIGHT
- DEMOLITION WORK LINEWEIGHT
- AREA OF DEMOLITION
- EXISTING WORK LINEWEIGHT
- FUTURE WORK LINEWEIGHT
- POINT OF CONNECTION
- TERMINATION POINT OF DEMOLITION
- POINT OF FUTURE CONNECTION
- # SHEET NOTE REFERENCE
- # REVISION REFERENCE
- E-1 EQUIPMENT TAG
- NORTH ARROW

CD-1 / 8" / 150 — NECK: CD-1 / 150; TYPE: CD-1-150 — DIFFUSER, REGISTER, OR GRILLE DESIGNATION AND AIRFLOW TAG(S); CFM

VAV-1 / 8" / 150 — DESIGNATION; DUCT INLET/PIPE SIZE; CFM/GPM — VAV/AIR VALVE TAG

1 VIEW TITLE — INDICATES PLAN/DETAIL NUMBER; PLAN/DETAIL TITLE; SCALE: SCALE (AS REQUIRED)

1 / M301 — INDICATES PLAN/DETAIL NUMBER; INDICATES DRAWING ON WHICH DRAWING SECTION APPEARS

PIPING

SYMBOL	DESCRIPTION
	PIPING TURNING UP
	PIPING TURNING DOWN
	TEE DOWN
	PIPE BRANCH BOTTOM TAKEOFF
	PIPE BRANCH TOP TAKEOFF
	VALVE IN VERTICAL PIPE
	CAP
	FLOW DIRECTION
R/D	PIPE SLOPE RISE/DROP
	PIPE RISER UP/DOWN
	CONCENTRIC REDUCER/INCREASER
	ECCENTRIC REDUCER/INCREASER (FOB)
	UNION
	PIPE CONTINUATION
W H	EXPANSION LOOP (WxH)
	PIPE ANCHOR
	FLEXIBLE PIPE CONNECTOR
EJ	EXPANSION JOINT
	PIPE GUIDE
	STEAM TRAP
	BLIND FLANGE
T	TEMPERATURE SENSOR
P	PRESSURE SENSOR
DP	DIFFERENTIAL PRESSURE TRANSDUCER
FS	FLOW SWITCH
FM	FLOW METER
	VENTURI FLOW METER
	PUMP

DUCTWORK

SYMBOL DOUBLE LINE	DESCRIPTION
20x12	DUCTWORK, SIZE AS INDICATED
20x12	DUCTWORK, WITH SOUNDLINING
	VANED ELBOW
	UN-VANED ELBOW
	RADIUS ELBOW
	VANED TEE
	UN-VANED TEE
	SUPPLY DUCT TURNING UP AND DOWN
	RETURN DUCT TURNING UP AND DOWN
	EXHAUST/RELIEF DUCT TURNING UP AND DOWN
R/D	DUCT RISE/DROP IN AIRFLOW DIRECTION
	DUCT TRANSITION, SQUARE TO SQUARE
	DUCT TRANSITION, SQUARE TO ROUND
	DUCT CAPPED
	45 DEGREE TAP
	CONICAL TAP
24V	SPUN-IN RUNOUT FITTING W/ FLEX. & VD
	FLEXIBLE DUCT
	FLEXIBLE CONNECTION
	MANUAL VOLUME DAMPER
	FIRE DAMPER
	SMOKE DAMPER
	COMBINATION FIRE/SMOKE DAMPER
BDD	BACKDRAFT DAMPER
MD	MOTORIZED DAMPER
BD	BAROMETRIC DAMPER
	DUCT CONTINUATION
	SUPPLY AIR FLOW DIRECTION
OA	OUTSIDE AIR FLOW DIRECTION
	RETURN/EXHAUST AIR FLOW DIRECTION
	RETURN, EXHAUST GRILLE OR REGISTER
	SUPPLY AIR DIFFUSER, 4-WAY BLOW
	SUPPLY AIR DIFFUSER, 3-WAY BLOW
	SUPPLY AIR DIFFUSER, 2-WAY BLOW
	SUPPLY AIR DIFFUSER, 1-WAY BLOW
	LINEAR SLOT DIFFUSER
UC	UNDERCUT DOOR
DL 12x12	DOOR LOUVER
AD	DUCT ACCESS DOOR
T	THERMOSTAT, MASTER THERMOSTAT
TS	TEMPERATURE SENSOR
H	HUMIDISTAT
FZ	FREEZESTAT
HS	HUMIDITY SENSOR
H	DUCT SMOKE DETECTOR
CO2	CARBON DIOXIDE SENSOR
CO	CARBON MONOXIDE SENSOR
NOx	NITROGEN OXIDES SENSOR
SOx	SULFUR OXIDES SENSOR
	GAS CONCENTRATION MONITOR
SP	STATIC PRESSURE SENSOR
AFMS	AIR FLOW MEASURING STATION
S	SWITCH OR TOGGLE SWITCH
TM	MANUAL TIMER SWITCH
HOA	HAND/OFF/AUTO SWITCH
1 2	STATIC PRESSURE CLASSIFICATION DESIGNATION (IN. W.G.)
S	SPEED SELECTOR

MECH VALVES

SYMBOL	THREE LINE PLAN	THREE LINE ELEVATION	DESCRIPTION
			GEN. SHUTOFF VALVE (BALL, GATE, BUTTERFLY)
			GATE VALVE
			GLOBE VALVE
			OUTSIDE STEM & YOKE VALVE
			BALL VALVE
			BUTTERFLY VALVE
			CHECK VALVE
			GAS COCK
			PRESSURE REDUCING VALVE
			CALIBRATED BALANCING VALVE
			CONTROL VALVE, TWO-WAY (ELECTRIC)
			CONTROL VALVE, THREE-WAY (ELECTRIC)
			THREE-WAY MANUAL VALVE
			CONTROL VALVE, TWO-WAY (PNEUMATIC)
S			SOLENOID VALVE
			RELIEF/SAFETY VALVE
			STRAINER
			STRAINER WITH DRAIN VALVE AND CAP
H			PRESSURE GAUGE WITH COCK
T			THERMOMETER
			CONTROL VALVE, THREE-WAY (PNEUMATIC)
			FUSOMATIC GATE VALVE
AAV			AUTOMATIC AIR VENT (EXTEND TO DRAIN)
MAV			MANUAL AIR VENT
			PRESSURE/TEMPERATURE TEST PORT
			AQUASTAT
			NORMALLY CLOSED PORT (NC)

INSULATION THICKNESS SCHEDULE

FLUID OPERATING TEMPERATURE RANGE AND USAGE (F)	INSULATION CONDUCTIVITY		NOMINAL PIPE OR TUBE SIZE (INCHES)				
	CONDUCTIVITY (BTU-IN/(H-FT^2-F))	MEAN RATING TEMP. (F)	< 1	1 TO < 1.5	1.5 TO < 4	4 TO < 8	≥ 8
HEATING SYSTEMS (HOT WATER, STEAM CONDENSATE)							
141 - 200	0.25 - 0.29	125	1.5	1.5	2	2	2
105 - 140	0.21 - 0.28	100	1	1	1.5	1.5	1.5
COOLING SYSTEMS (CHILLED WATER, REFRIGERANT, AIR CONDITIONING CONDENSATE)							
40 - 60	0.21 - 0.27	75	0.5	0.5	1	1	1
< 40	0.20 - 0.26	50	0.5	1	1	1	1.5

DUCTWORK	INSULATION TYPE				JACKET
	GLASS FIBER WRAP	CLOSED CELL WRAP	CLOSED CELL LINING	FOAM BOARD (POLYISO)	
INDOORS: SUPPLY & RETURN DUCTWORK & PLENUMS IN UNCONDITIONED SPACES (R-6 INSTALLED)	2"	1.5" TYPE FS	-	-	-

PIPING MATERIAL SCHEDULE

SERVICE	SIZE (IN)	APPROVED MATERIAL
DUAL TEMP WATER HEATING HOT WATER CHILLED WATER	3" Ø OR SMALLER	SOLDERED PRESS SEAL, OR BRAZED TYPE 'L' COPPER THREADED WELDED OR GROOVED JOINTS SCHEDULE 40 CARBON STEEL
DUAL TEMP WATER HEATING HOT WATER CHILLED WATER	GREATER THAN 3"	BRAZED TYPE 'L' COPPER WELDED OR GROOVED SCHEDULE 40 CARBON STEEL
CONDENSATE DRAIN	ALL SIZES	SOLDERED TYPE 'M' COPPER
NATURAL GAS ≤ 14 W.C	4" Ø OR SMALLER	THREADED SCHEDULE 40 BLACK STEEL
NATURAL GAS	GREATER THAN 4"	WELDED SCHEDULE 40 BLACK STEEL
NATURAL GAS > 14 W.C	ALL SIZES	WELDED SCHEDULE 40 BLACK STEEL

CONDENSATE DRAIN SIZING

EQUIPMENT CAPACITY	MINIMUM CONDENSATE PIPE DIAMETER
UP TO 20 TONS OF REFRIGERATION	3/4 INCH

GANNETT FLEMING

Gannett Fleming, Inc.

1010 Adams Avenue
Valley Forge • PA • 19403
t 610 • 650 • 8156

www.gannettfleming.com

READING FIRE DEPARTMENT, ENGINE 7

950 MCKNIGHT ST, READING, PA 19601

HVAC REPLACEMENT / IAQ UPGRADE

NO.	DATE	DESCRIPTION

REVISIONS

BID DOCUMENTS

PROJECT NUMBER: 066446.007

DESIGNED:	MODELED:	CHECKED:
SAJ	AMF	MJM

DATE: 22 JULY 2022

DRAWING TITLE: MECHANICAL COVER SHEET

DRAWING NO.: SHEET 2 OF 17

M001

Local File:BIM 360:///066446-City_of_Reading/066446_007_MEP_Central_R21.rvt

Project Number: 066446.007
7/22/2022 7:17:11 PM

ABBREVIATIONS

AAV	AUTOMATIC AIR VENT
ABV	ABOVE
ACCU	AIR-COOLED CONDENSING UNIT
AD	AREA DRAIN, ACCESS DOOR, AIR DRYER
ADA	AMERICANS WITH DISABILITIES ACT
ADJ	ADJUST, ADJUSTABLE
AF	ANTI-FREEZE
AFF	ABOVE FINISHED FLOOR
AHU	AIR-HANDLING UNIT
ALUM	ALUMINUM
AP	ACCESS PANEL
APPROX	APPROXIMATELY
ARCH	ARCHITECTURAL
AS	AIR SEPARATOR
AST	ABOVEGROUND STORAGE TANK
ATV	ATMOSPHERIC VENT
AVG	AVERAGE
BA	BREATHABLE AIR
BAS	BUILDING AUTOMATION SYSTEM
BFF	BELOW FINISHED FLOOR
BFP	BACKFLOW PREVENTER
BLDG	BUILDING
BLW	BELOW
BMS	BUILDING MANAGEMENT SYSTEM
BOD	BASIS OF DESIGN
BOT	BOTTOM
BTU	BRITISH THERMAL UNIT
BTUH	BRITISH THERMAL UNITS PER HOUR
C	COMMON
CA	COMPRESSED AIR
CD	CONDENSATE DRAIN
CFH	CUBIC FEET PER HOUR
CHWR	CHILLED WATER RETURN
CHWS	CHILLED WATER SUPPLY
CI	CAST IRON
CL	CENTER LINE
CLDIP	CEMENT-LINED DUCTILE IRON S PIPE
CLNG	CEILING
CNG	COMPRESSED NATURAL GAS
CO	CLEANOUT, CARBON MONOXIDE
CO2	CARBON DIOXIDE
COL	COLUMN
CONC	CONCRETE
CONN	CONNECTION
CONT	CONTINUATION
CONT'D	CONTINUED
CPVC	CHLORINATED POLYVINYL CHLORIDE
CV	CLEANING VACUUM, CHECK VALVE
CVC	CONVECTOR
CW	COLD WATER
CWR	CONDENSER WATER RETURN
CWS	CONDENSER WATER SUPPLY
Cx	COMMISSIONING
CxA	COMMISSIONING AUTHORITY
(D)	DEMOLISH, REMOVE
D	DRAIN, DEPTH
DB	DECIBELS
DCW	DOMESTIC COLD WATER
DDC	DIRECT DIGITAL CONTROL
DEG, °	DEGREES
DEH	DEHUMIDIFIER
DEPT	DEPARTMENT
DESG	DESIGNATION
DFU	DRAINAGE FIXTURE UNITS
DHW	DOMESTIC HOT WATER
DHWR	DOMESTIC HOT WATER RETURN
DI	DEIONIZED WATER
DIA, ø	DIAMETER
DIFF	DIFFERENTIAL, DIFFERENCE
DISC	DISCONNECT
DN	DOWN
DSB	DOWNSPOUT BOOT
DSN	DOWNSPOUT NOZZLE
DTWR	HOT/CHILLED WATER RETURN (DUAL TEMP.)
DTWS	HOT/CHILLED WATER SUPPLY (DUAL TEMP.)
DWG	DRAWING
(ER)	EXISTING RELOCATED
(EX)	EXISTING TO REMAIN
EA	EACH
EAT	ENTERING AIR TEMPERATURE
EFF	EFFICIENCY
EJ	EXPANSION JOINT
EL	ELEVATION
ELEC	ELECTRIC, ELECTRICAL
ELEV	ELEVATOR
EQ	EQUAL
ET	EXPANSION TANK
EWC	ELECTRIC WATER COOLER
EWT	ENTERING WATER TEMPERATURE
EXT	EXTERIOR
(F)	FUTURE
FCO	FLOOR CLEANOUT
FD	FLOOR DRAIN, FIRE DAMPER
FDC	FIRE DEPARTMENT CONNECTION
FDR	FLOOR DRAIN
FF	FINISHED FLOOR
FG	FUEL GAS
FLA	FULL LOAD AMPS
FLEX	FLEXIBLE
FLR	FLOOR
FM	FLOW METER
FO	FUEL OIL
FOF	FUEL OIL FILL
FOR	FUEL OIL RETURN
FOS	FUEL OIL SUPPLY
FOV	FUEL OIL VENT
FPM	FEET PER MINUTE
FPS	FEET PER SECOND
FRP	FIBERGLASS REINFORCED PLASTIC
FS	FLOW SWITCH
FT HD	FEET OF HEAD
FT., '	FOOT, FEET
FZ	FREEZE, FREEZSTAT
°F	DEGREES FAHRENHEIT
G	GAS
GA	GAUGE
GAL	GALLON, GALLONS
GALV	GALVANIZED
GCO	GRADE CLEANOUT
GEN	GENERATOR
GI	GREASE INTERCEPTOR
GPH	GALLONS PER HOUR
GPM	GALLONS PER MINUTE
GW	GREASE WASTE
GWR	GLYCOL WATER SUPPLY
GWS	GLYCOL WATER RETURN
H	HEIGHT
H2O	WATER
HB	HOSE BIB
HHWR	HEATING HOT WATER RETURN
HHWS	HEATING HOT WATER SUPPLY
HORIZ	HORIZONTAL
HP	HORSEPOWER
HPS	HIGH PRESSURE STEAM
HTG	HEATING
HUM	HUMIDIFIER
HWR	HOT WATER RETURN
HWS	HOT WATER SUPPLY
HX	HEAT EXCHANGER
HZ	HERTZ (CYCLES PER SECOND)
ID	INSIDE DIAMETER
IN WG	INCHES OF WATER GAUGE
IN, "	INCH, INCHES
INFO	INFORMATION
INFO	INFORMATION
INT	INTERIOR
INT	INTERIOR
INV EL	INVERT ELEVATION
ISOL	ISOLATION
IW	INDUSTRIAL WASTE, INDIRECT WASTE
KW	KILOWATTS
L	LENGTH, LONG
LAT	LEAVING AIR TEMPERATURE
LAV	LAVATORY
LBS	POUNDS
LDR	LEADER
LP	LIQUID PROPANE
LPCR	LOW PRESSURE CONDENSATE RETURN
LPS	LOW PRESSURE STEAM
LRA	LOAD RATED AMPS
LWT	LEAVING WATER TEMPERATURE
M	METER
MA	MEDICAL AIR
MAV	MANUAL AIR VENT
MAX	MAXIMUM
MBH	THOUSAND BRITISH THERMAL UNITS PER HOUR
MCA	MINIMUM CIRCUIT AMPACITY
MECH	MECHANICAL
MFR	MANUFACTURER
MIN	MINIMUM
MO	MOTOR OIL
MOCP	MAXIMUM OVERCURRENT PROTECTION
MPS	MEDIUM PRESSURE STEAM
MTD	MOUNTED
MTG	MOUNTING
MUW	MAKE UP WATER
MV	MEDICAL VACUUM
N	NITROGEN
N/A	NOT APPLICABLE
NC	NORMALLY CLOSED, NOISE CRITERIA
NFPA	NATIONAL FIRE PROTECTION ASSOCIATION
NFWH	NON-FREEZE WALL HYDRANT
NG	NATURAL GAS
NIC	NOT IN CONTRACT
NO	NORMALLY OPENED
No	NUMBER
NOM	NOMINAL
NPW	NON-POTABLE WATER
NTS	NOT TO SCALE
O/2, O2	OXYGEN
OC	ON CENTER
OFD	OVERFLOW DRAIN
OI	OIL INTERCEPTOR
OS&Y	OUTSIDE STEM AND YOKE VALVE
OSD	OPEN SITE DRAIN
OVH	OVERHEAD
OWS	OIL WATER SEPARATOR
P	PUMP
PC	PUMPED CONDENSATE
PD	PRESSURE DROP
PH	PHASE
PIW	PUMPED INDUSTRIAL WASTE
PLBG	PLUMBING
PNEU	PNEUMATIC
POS	POSITION
PRESS	PRESSURE
PROP	PROPELLER
PRV	PRESSURE REDUCING VALVE
PS	PRESSURE SWITCH
PSC	PUMPED STEAM CONDENSATE RETURN
PSI	POUNDS PER SQUARE INCH
PSIG	POUNDS PER SQUARE INCH GAUGE
PVC	POLYVINYL CHLORIDE
(RL)	RELOCATED POSITION
R	RISE
RD	ROOF DRAIN, REFRIGERANT DISCHARGE
REF	REFRIGERANT
REQ'D	REQUIRED
RL	REFRIGERANT LIQUID
RLA	RATED LOAD AMPS
RM	ROOM
RPM	REVOLUTIONS PER MINUTE
RPZ	REDUCED PRESSURE BFP
RS	REFRIGERANT SUCTION
RWC	RAIN WATER CONDUCTOR
S	SWITCH
S/S	START/STOP
SAN	SANITARY
SC	STEAM CONDENSATE RETURN
SCCR	SHORT-CIRCUIT CURRENT RATING
SF	SUPPLY FAN, SQUARE FEET
SK	SINK
SP	STATIC PRESSURE
SPEC	SPECIFICATION
SPR	SPRINKLER
SQ	SQUARE
SQ FT	SQUARE FOOT, SQUARE FEET
SS	STAINLESS STEEL
ST	STORM WATER PIPING
STD	STANDARD
STL	STEEL
STM	STEAM
TEMP	TEMPERATURE, TEMPORARY
TP	TRAP PRIMER
TW	TEPID WATER
TYP	TYPICAL
UH	UNIT HEATER
UL	UNDERWRITERS LABORATORIES
UON	UNLESS OTHERWISE NOTED
UV	UNIT VENTILATOR
V	VOLTS, VENT
VCO	VERTICAL CLEAN OUT
VEL	VELOCITY
VERT	VERTICAL
VFD	VARIABLE FREQUENCY DRIVE
VIB	VIBRATION
VSD	VARIABLE SPEED DRIVE
VTR	VENT THROUGH ROOF
W	WIDTH, WIDE
W/	WITH
W/O	WITHOUT
WAGD	WASTE ANESTHETIC GAS DISPOSAL
WC	WATER CLOSET
WCO	WALL CLEANOUT
WFU	WATER SUPPLY FIXTURE UNITS
WG	WATER GAUGE
WH	WATER HEATER, WALL HYDRANT
WHA	WATER HAMMER ARRESTOR
WMS	WIRE MESH SCREEN
WPD	WATER PRESSURE DROP
WSHP	WATER SOURCE HEAT PUMP
WT	WEIGHT
ZN	ZONE

GENERAL NOTES

1. NOT ALL ABBREVIATIONS, LINE TYPES, OR SYMBOLS MAY APPEAR ON THESE CONTRACT DOCUMENTS.
2. DRAWINGS SHALL BE CONSIDERED DIAGRAMMATIC AND ARE INTENDED TO INDICATE CAPACITY, SIZE, APPROXIMATE LOCATION AND GENERAL ARRANGEMENT. WHILE THE DRAWINGS ARE GENERALLY TO SCALE AND ARE AS ACCURATE AS THE SCALE WILL PERMIT, DIMENSIONS SHALL BE CONFIRMED IN THE FIELD.
3. THE CONTRACTOR SHALL COMPLY WITH THE LAWS, ORDINANCES, RULES AND REGULATIONS OF LOCAL AND STATE GOVERNMENTAL AUTHORITIES; OF THE NATIONAL FIRE PROTECTION ASSOCIATION AS INTERPRETED BY THE ENFORCING AUTHORITY HAVING JURISDICTION; AND OF PUBLIC UTILITIES HAVING CONNECTION WITH ANY OF THE SYSTEMS HEREIN SPECIFIED.
4. INSTALL EQUIPMENT IN A SERVICEABLE MANNER WITHOUT DISRUPTION TO ADJACENT SERVICES OR DAMAGING INSULATION. THIS INCLUDES, BUT IS NOT LIMITED TO: PROVIDE VALVE OPERATOR EXTENSIONS TO EXTEND THRU THE INSULATION AND HANDLES TO SWING TOWARD THE OPERATOR FOR ISOLATION.
5. PROVIDE ALL NEW AND REUSED WALL PENETRATIONS WITH A SLEEVE/CONDUIT FOR SERVICES PENETRATING IT. THIS INCLUDES, BUT IS NOT LIMITED TO: DUCTWORK, PIPING, POWER, AND CONTROL WIRING.
6. THE CONTRACTOR SHALL SECURE AND PAY FOR ALL PERMITS AND INSPECTIONS REQUIRED BY ANY OF THE FOREGOING AUTHORITIES, AND PAY FOR ALL OTHER COSTS IN CONNECTION WITH THE WORK. CERTIFICATES SHALL BE IN DUPLICATE AND SHALL BE DELIVERED TO THE OWNER.
7. THE SITE, LOCATION, AND ROUTING OF SYSTEMS INDICATED TO HAVE NEW CONNECTIONS MADE TO THEM ARE SHOWN AS ACCURATELY AS FIELD CONDITIONS PERMIT. CONTRACTOR SHALL VISIT THE SITE AND THOROUGHLY EXAMINE THE CONTRACT DRAWINGS. ALL EXISTING CONDITIONS SHALL BE EXAMINED AND THEIR EXACT LOCATIONS VERIFIED. THE CONTRACTOR SHALL REPORT TO THE ENGINEER ANY CONDITIONS WHICH MIGHT MAKE INSTALLATION OF REQUIRED EQUIPMENT A PROBLEM. NO CONSIDERATION OR ALLOWANCE WILL BE GRANTED FOR FAILURE TO INVESTIGATE CONDITIONS OR MISUNDERSTANDINGS OF THE CONTRACTUAL REQUIREMENTS.
8. THE CONTRACTOR SHALL INSTALL AND CONNECT EQUIPMENT AND MATERIALS IN ACCORDANCE WITH THE BEST ENGINEERING PRACTICE AND, UNLESS OTHERWISE SHOWN OR SPECIFIED, FOLLOW THE MANUFACTURER'S PRINTED INSTALLATION REQUIREMENTS AND RECOMMENDATIONS, AND FURNISH AND INSTALL REQUIRED AUXILIARY ITEMS TO PROVIDE A COMPLETE INSTALLATION.
9. THE CONTRACTOR SHALL REPAIR WALLS, CEILING, FLOORS, ETC., THAT ARE REQUIRED TO BE PENETRATED, OR OTHERWISE DISTURBED. THE REPAIRS SHALL BE WITH MATERIALS AND FINISHES TO MATCH EXISTING. FIRE WALL PENETRATIONS SHALL BE SEALED WITH SUITABLE MATERIALS TO PRESERVE FIRE WALL INTEGRITY.
10. THE CONTRACTOR SHALL REMOVE EQUIPMENT NOT INDICATED TO BE REUSED TO A DESIGNATED LOCATION AT THE PROJECT SITE. AFTER THE EQUIPMENT HAS BEEN ASSEMBLED FOR THE OWNER'S INSPECTION AND POSSIBLE RETENTION, ALL EQUIPMENT NOT TO BE RETAINED BY THE OWNER SHALL BE REMOVED FROM THE SITE BY THE CONTRACTOR.
11. COORDINATE WITH OTHER TRADES TO AVOID INTERFERENCE AMONG MECHANICAL, ELECTRICAL, ARCHITECTURAL AND STRUCTURAL, ETC. PROVIDE NECESSARY OFFSETS IN PIPING, DUCTWORK AND FITTINGS ETC., REQUIRED TO PROPERLY INSTALL WORK WITHOUT INTERFERENCES.
12. BUILDING SYSTEMS SHALL REMAIN IN SERVICE UNLESS INDICATED OTHERWISE. OUTAGES OR INTERRUPTIONS SHALL BE KEPT TO A MINIMUM DURATION. NOTIFY THE OWNER 48 HOURS IN ADVANCE OF ANY OUTAGE OR INTERRUPTION. IF TEMPORARY CONNECTIONS ARE NECESSARY TO ASSURE THIS CONTINUITY OF SERVICES, THEY SHALL BE FURNISHED BY THE CONTRACTOR WITHOUT ADDITIONAL CHARGE TO THE OWNER AND SHALL BE REMOVED WHEN NO LONGER NEEDED.
13. THE CONTRACTOR SHALL ARRANGE AND EXECUTE HIS WORK SUCH THAT ANY CONNECTIONS, BOTH TEMPORARY OR PERMANENT, TO, OR REARRANGEMENT OF, PRESENT EQUIPMENT, PIPING, ETC., SHALL BE MADE IN SUCH A MANNER AS TO ASSURE FULL RESUMPTION OF SERVICE AT THE TIME DESIGNATED BY THE OWNER.
14. THE CONTRACTOR SHALL LOCATE EQUIPMENT WHICH MUST BE SERVICED, OPERATED OR MAINTAINED IN FULLY ACCESSIBLE POSITIONS. FURNISH ACCESS DOORS AS REQUIRED FOR BETTER ACCESSIBILITY. MINOR DEVIATIONS FROM DRAWINGS MAY BE MADE TO ALLOW FOR ACCESSIBILITY, BUT CHANGES OF MAGNITUDE WHICH INVOLVE EXTRA COSTS SHALL NOT BE MADE WITHOUT WRITTEN APPROVAL.
15. ALL DIMENSIONS FOR EXISTING PIPING AND DUCTWORK SHALL BE VERIFIED IN THE FIELD BY THE CONTRACTOR PRIOR TO CONSTRUCTION, WHERE NEW WORK CONNECTIONS ARE INDICATED. NEW DUCT DIMENSIONS SHOWN ARE INSIDE CLEAR DIMENSIONS.
16. OPENINGS REMAINING IN EXISTING PIPING/DUCTWORK AS A RESULT OF DEMOLITION SHALL BE SEALED WITH AN AIRTIGHT/WATERTIGHT SHEET METAL CAP. WHERE EXISTING SYSTEMS ARE INSULATED, WORK SHALL INCLUDE REPAIR AND REPLACEMENT OF INSULATION EITHER DAMAGED OR REMOVED AS A RESULT OF DEMOLITION.
17. PIPING SCOPE REQUIREMENTS: HORIZONTAL DRAINAGE PIPING SHALL BE INSTALLED IN UNIFORM ALIGNMENT AT UNIFORM SLOPES. THE MINIMUM SLOPE OF HORIZONTAL DRAINAGE PIPE SHALL BE 1/4" PER FOOT.
18. ALL REFRIGERANT SHALL BE RECOVERED AND CHARGED IN ACCORDANCE WITH THE MANUFACTURER'S REQUIREMENTS AND LOCAL EPA REQUIREMENTS.

GENERAL DEMOLITION NOTES

1. EXISTING HANGERS SHALL BE MODIFIED TO FACILITATE NEW ABOVE CEILING WORK.
2. EQUIPMENT NO LONGER IN SERVICE SHALL BE REMOVED IN ITS ENTIRETY AS REQUIRED TO FACILITATE NEW WORK.

GANNETT FLEMING

Gannett Fleming, Inc.

1010 Adams Avenue
Valley Forge • PA • 19403
t 610 • 650 • 8156

www.gannettfleming.com

READING FIRE DEPARTMENT, ENGINE 7

950 MCKNIGHT ST, READING, PA 19601

HVAC REPLACEMENT / IAQ UPGRADE

NO.	DATE	DESCRIPTION

REVISIONS

BID DOCUMENTS

PROJECT NUMBER: 066446.007

DESIGNED:	MODELED:	CHECKED:
SAJ	AMF	MJM

DATE: 22 JULY 2022

DRAWING TITLE: MECHANICAL

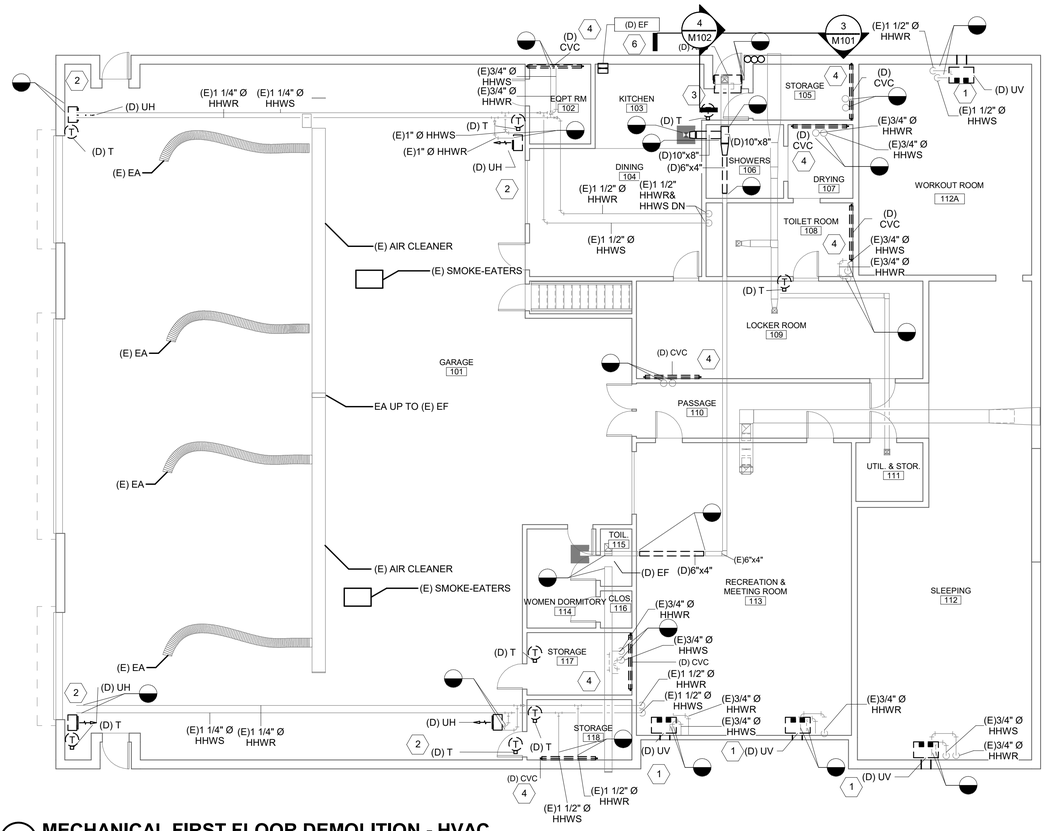
GENERAL NOTES AND ABBREVIATIONS

DRAWING NO.: SHEET 3 OF 17

M002

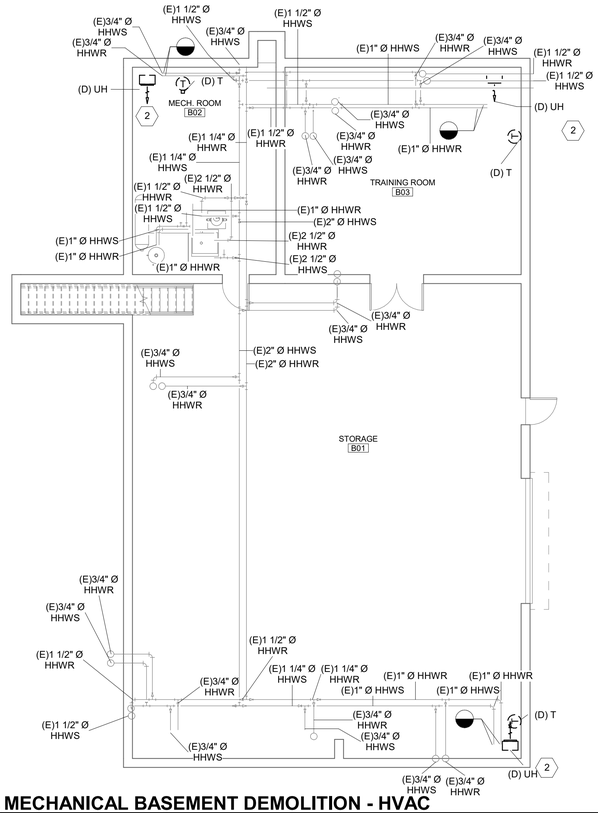
Local File:BIM 360://066446-City_of_Reading/066446_007_MEP_Central_R21.rvt

Project Number: 066446.007
7/22/2022 7:17:11 PM



1 MECHANICAL FIRST FLOOR DEMOLITION - HVAC

SCALE: 1/8" = 1'-0"



2 MECHANICAL BASEMENT DEMOLITION - HVAC

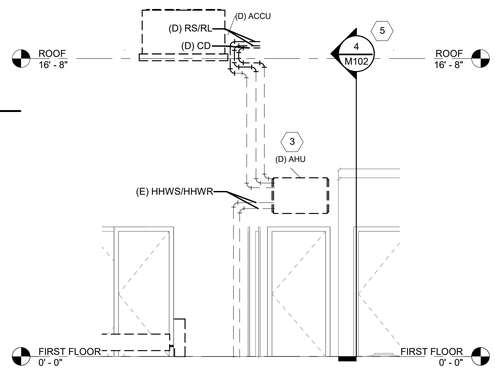
SCALE: 1/8" = 1'-0"

SHEET NOTES

1. PROVIDE CLEANING OF ALL EXISTING DUCTWORK TO REMAIN.
2. PROVIDE CUTTING, PATCHING, REPAIR, AND FINISHING OF ALL EXISTING SURFACES IMPACTED BY DEMOLITION TO MATCH EXISTING SURFACES.
3. ALL NEW CONTROL VALVES SHALL BE 3-WAY TYPE.

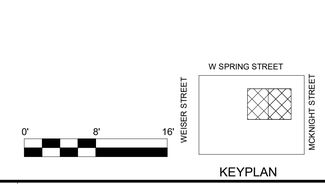
SHEET DEMOLITION NOTES

1. REMOVE EXISTING CABINET UNIT VENTILATOR AND ASSOCIATED SUPPORTS, ACCESSORIES, OUTSIDE AIR INTAKE, ELECTRICAL CONNECTION, CONTROLS, AND THERMOSTAT. REMOVE EXISTING HEATING HOT WATER SUPPLY AND RETURN PIPING CONNECTIONS AND TEMPORARILY CAP. REMOVE EXISTING CONTROL VALVE. CLEAN AND INSULATE OUTDOOR AIR OPENING.
2. REMOVE EXISTING HYDRONIC UNIT HEATER AND ASSOCIATED SUPPORTS, ACCESSORIES, ELECTRICAL CONNECTIONS, CONTROLS, AND THERMOSTAT. REMOVE EXISTING HEATING HOT WATER SUPPLY AND RETURN PIPING CONNECTIONS AND TEMPORARILY CAP. REMOVE EXISTING CONTROL VALVE.
3. REMOVE EXISTING CEILING-MOUNTED AIR HANDLING UNIT AND ASSOCIATED SUPPORTS, ACCESSORIES, ELECTRICAL CONNECTIONS, CONTROLS. PIPING AND DUCTWORK CONNECTIONS, AND THERMOSTAT. REMOVE EXISTING HEATING HOT WATER SUPPLY AND RETURN PIPING CONNECTIONS AND TEMPORARILY CAP. REMOVE EXISTING CONTROL VALVE. REMOVE REFRIGERANT PIPING AND SUPPORTS UP TO CONDENSING UNIT. TEMPORARILY SEAL ROOF PIPING PENETRATION. PREPARE FOR REUSE.
4. REMOVE EXISTING CONVECTOR AND ASSOCIATED SUPPORTS, ACCESSORIES, CONTROLS, PIPING CONNECTIONS, AND THERMOSTAT. REMOVE EXISTING HEATING HOT WATER SUPPLY AND RETURN PIPING CONNECTIONS AND TEMPORARILY CAP.
5. REMOVE EXISTING AIR-COOLED CONDENSING UNIT, ASSOCIATED SUPPORTS, ACCESSORIES, ELECTRICAL CONNECTIONS, CONTROLS, AND PIPING CONNECTIONS. PREPARE EQUIPMENT SUPPORTS FOR REUSE.
6. REMOVE EXISTING EXHAUST FAN AND ASSOCIATED SUPPORTS, ACCESSORIES, ELECTRICAL CONNECTIONS AND CONTROLS.



3 AHU-1 SECTION DEMOLITON - HVAC

SCALE: 1/4" = 1'-0"



GANNETT FLEMING

Gannett Fleming, Inc.

1010 Adams Avenue
Valley Forge • PA • 19403
t 610 • 650 • 8156

www.gannettfleming.com

READING FIRE DEPARTMENT, ENGINE 7

950 MCKNIGHT ST, READING, PA 19601

HVAC REPLACEMENT / IAQ UPGRADE

NO.	DATE	DESCRIPTION

REVISIONS

BID DOCUMENTS

PROJECT NUMBER: 066446.007

DESIGNED:	MODELED:	CHECKED:
SAJ	AMF	MJM

DATE: 22 JULY 2022

DRAWING TITLE: MECHANICAL

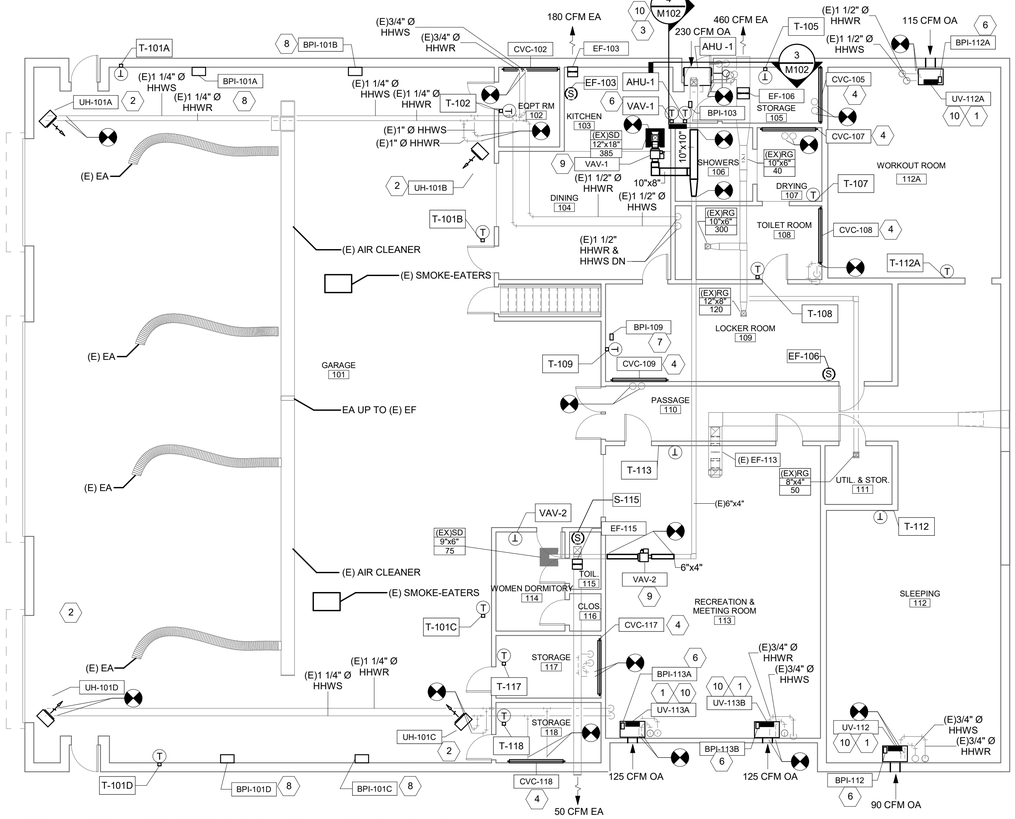
DEMOLITION BASEMENT AND FIRST FLOOR - HVAC

DRAWING NO.: SHEET 4 OF 17

M101

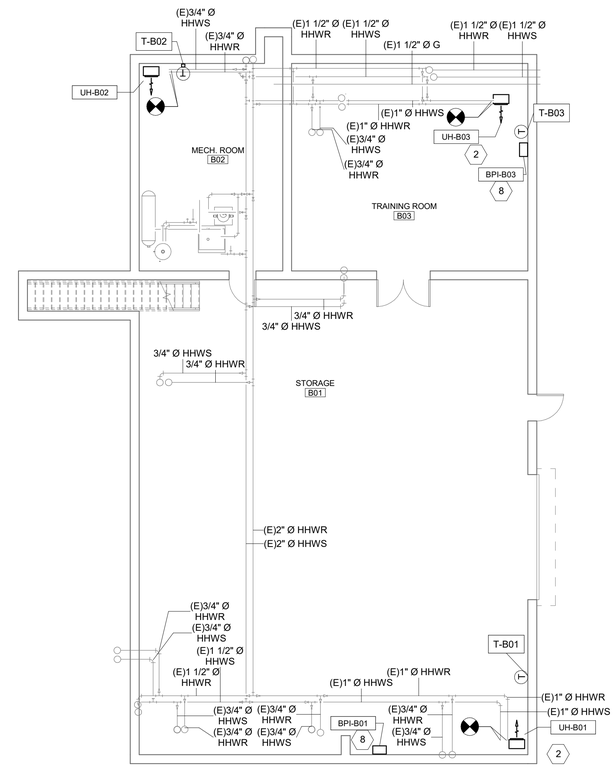
Local File:BIM 360://066446-City_of_Reading/066446_007_MEP_Central_R21.rvt

Project Number: 066446.007
7/22/2022 7:17:13 PM



1 MECHANICAL FIRST FLOOR NEW WORK - HVAC

SCALE: 1/8" = 1'-0"



2 MECHANICAL BASEMENT NEW WORK - HVAC

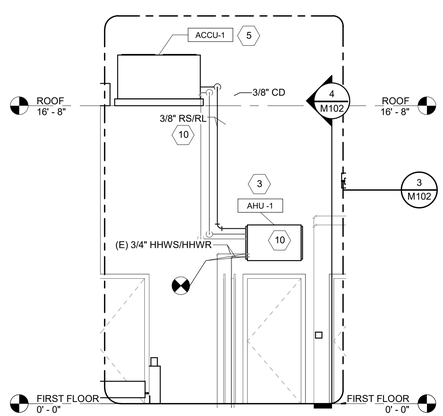
SCALE: 1/8" = 1'-0"

SHEET NEW WORK NOTES

1. PROVIDE NEW CABINET UNIT VENTILATOR ACCESSORIES, AND WALL-MOUNTED REMOTE, PROGRAMMABLE THERMOSTAT. PROVIDE OUTSIDE AIR OPENING AND OUTSIDE AIR DAMPER. CONNECT EXISTING HEATING HOT WATER SUPPLY AND RETURN PIPING. CONNECT TO NEW REFRIGERANT PIPING. THERMOSTAT LOCATION TO BE COORDINATED WITH OWNER.
2. PROVIDE NEW HYDRONIC UNIT HEATER, SUPPORT, ACCESSORIES AND WALL-MOUNTED THERMOSTAT WITH ON/OFF/AUTO SWITCH. CONNECT EXISTING HEATING HOT WATER SUPPLY AND RETURN PIPING TO HYDRONIC UNIT HEATER. THERMOSTAT LOCATION TO BE COORDINATED WITH OWNER.
3. PROVIDE NEW CEILING-MOUNTED AIR HANDLING UNIT, ACCESSORIES, AND WALL-MOUNTED THERMOSTAT. CONNECT EXISTING HEATING HOT WATER SUPPLY AND RETURN PIPING TO AIR HANDLING UNIT. CONNECT AIR HANDLING UNIT TO EXISTING SUPPLY AIR DUCTWORK. THERMOSTAT LOCATION TO BE COORINATED WITH THE OWNER. CONNECT TO NEW REFRIGERANT PIPING.
4. PROVIDE NEW CONVECTOR, ACCESSORIES, AND WALL-MOUNTED THERMOSTAT WITH ON/OFF/AUTO SWITCH. CONNECT EXISTING HEATING HOT WATER SUPPLY AND RETURN PIPING TO CONVECTOR. THERMOSTAT LOCATION TO BE COORDINATED WITH OWNER.
5. PROVIDE NEW AIR-COOLED CONDENSING ON ROOF. PLACE NEW UNIT ON EXISTING EQUIPMENT SUPPORT. CONNECT EXISTING REFRIGERANT PIPING TO CONDENSING UNIT.
6. PROVIDE NEW APPROPRIATE BIPOLAR IONIZATION EQUIPMENT PER EQUIPMENT SCHEDULE. SEE DETAIL FOR INSTALLATION INSTRUCTIONS.
7. PROVIDE NEW APPROPRIATE STANDALONE BIPOLAR IONIZATION EQUIPMENT PER EQUIPMENT SCHEDULE. PROVIDE TWO (2) WALL-MOUNTED 'L'-BRACKET SUPPORTS FOR BIPOLAR IONIZATION EQUIPMENT TO SIT ON, 6 FEET ABOVE FINISHED FLOOR.
8. PROVIDE NEW APPROPRIATE STANDALONE BIPOLAR IONIZATION EQUIPMENT PER EQUIPMENT SCHEDULE. PROVIDE TWO (2) WALL-MOUNTED 'L'-BRACKET SUPPORTS FOR BIPOLAR IONIZATION EQUIPMENT TO SIT ON, 2 FEET ABOVE FINISHED FLOOR.
9. PROVIDE NEW VARIABLE AIR VOLUME TERMINAL UNIT, ASSOCIATED CONTROLS AND ACCESSORIES.
10. PROVIDE INSULATED REFRIGERANT PIPING UP TO NEW CONDENSING UNIT. SEAL ROOF PENETRATION WEATHERTIGHT. PROVIDE CONDENSATE PUMP POWERED BY AIR HANDLING UNIT AND INSULATED CONDENSATE PIPING UP TO ROOF. EXTERIOR REFRIGERANT PIPING SHALL BE INSULATED WITH FIELD-APPLIED PVC JACKET.

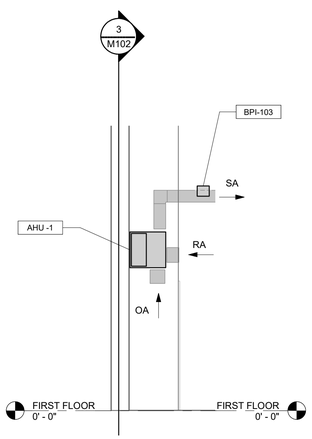
SHEET NOTES

1. PROVIDE CLEANING OF ALL EXISTING DUCTWORK TO REMAIN.
2. PROVIDE CUTTING, PATCHING, REPAIR, AND FINISHING OF ALL EXISTING SURFACES IMPACTED BY DEMOLITION TO MATCH EXISTING SURFACES.
3. ALL NEW CONTROL VALVES SHALL BE 3-WAY TYPE.



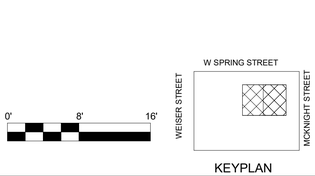
3 AHU-1 SECTION NEW WORK - HVAC PIPING

SCALE: 1/4" = 1'-0"



4 AHU-1 SECTION NEW WORK - HVAC DUCTWORK

SCALE: 1/4" = 1'-0"



KEYPLAN

GANNETT FLEMING

Gannett Fleming, Inc.

1010 Adams Avenue
Valley Forge • PA • 19403
t 610 • 650 • 8156

www.gannettfleming.com

READING FIRE DEPARTMENT, ENGINE 7

950 MCKNIGHT ST, READING, PA 19601

HVAC REPLACEMENT / IAQ UPGRADE

NO.	DATE	DESCRIPTION

REVISIONS

BID DOCUMENTS

PROJECT NUMBER: 066446.007

DESIGNED:	MODELED:	CHECKED:
SAJ	AMF	MJM

DATE: 22 JULY 2022

DRAWING TITLE: MECHANICAL

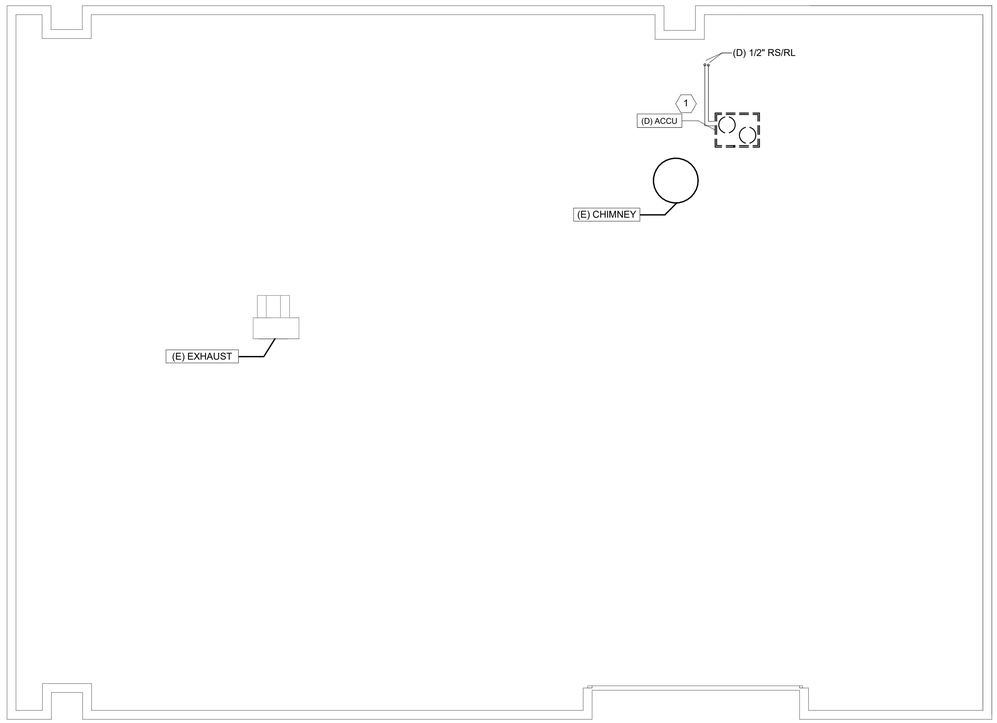
NEW WORK BASEMENT AND FIRST FLOOR - HVAC

DRAWING NO.: SHEET 5 OF 17

M102

Local File: BIM 360://066446-City_of_Reading/066446_007_MEP_Central_R21.rvt

Project Number: 066446.007
7/22/2022 7:17:15 PM



2 MECHANICAL DEMOLITON ROOF PLAN

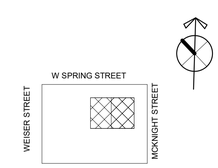
SCALE: 1/8" = 1'-0"

SHEET NOTES

1. COORDINATE ALL ROOFING WORK WITH THE OWNER.

SHEET DEMOLITION NOTES

1. REMOVE EXISTING AIR-COOLED CONDENSING UNIT, ASSOCIATED SUPPORTS, ACCESSORIES, ELECTRICAL CONNECTIONS, CONTROLS, AND PIPING CONNECTIONS. PREPARE EQUIPMENT SUPPORTS FOR REUSE.



KEYPLAN

GANNETT FLEMING

Gannett Fleming, Inc.

1010 Adams Avenue
Valley Forge • PA • 19403
t 610 • 650 • 8156

www.**gannettfleming**.com

READING FIRE DEPARTMENT, ENGINE 7

950 MCKNIGHT ST, READING, PA 19601

HVAC REPLACEMENT / IAQ UPGRADE

NO.	DATE	DESCRIPTION

REVISIONS

BID DOCUMENTS

PROJECT NUMBER: 066446.007

DESIGNED:	MODELED:	CHECKED:
AMF	AMF	MJM

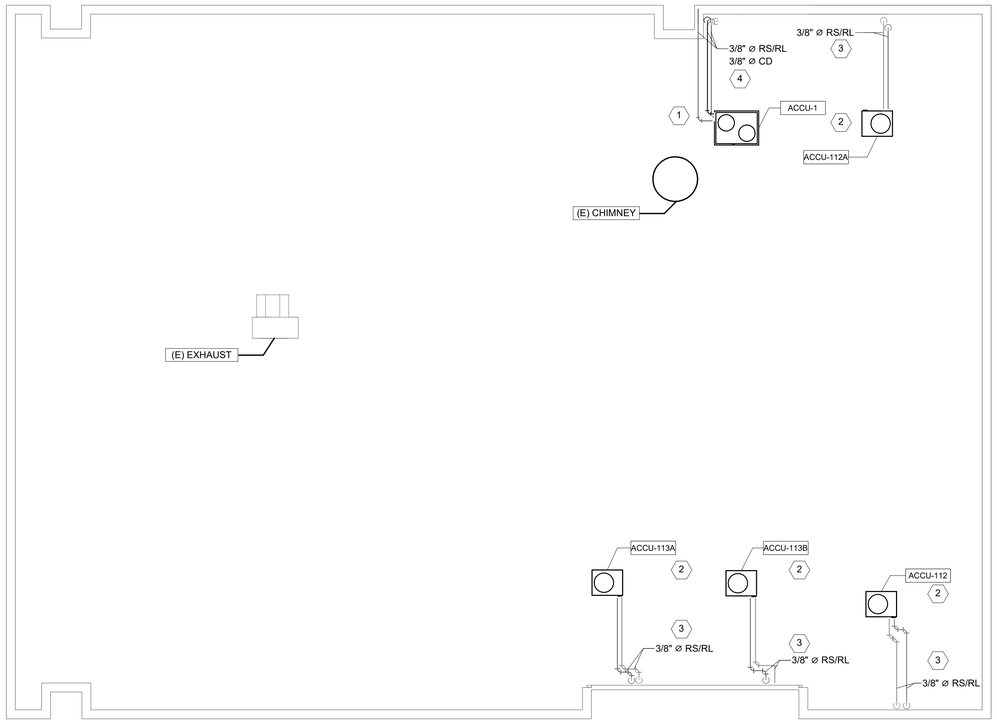
DATE: 22 JULY 2022

DRAWING TITLE: MECHANICAL DEMOLITON ROOF PLAN

DRAWING NO.: SHEET 6 OF 17

M103

Local File:BIM 360://066446-City_of_Reading/066446_007_MEP_Central_R21.rvt
Project Number: 066446.007
7/22/2022 7:17:15 PM



1 MECHANICAL NEW WORK ROOF PLAN

SCALE: 1/8" = 1'-0"

SHEET NOTES

1. COORDINATE ALL ROOFING WORK WITH THE OWNER.

SHEET NEW WORK NOTES

1. PROVIDE NEW AIR-COOLED CONDENSING ON ROOF. PLACE NEW UNIT ON EXISTING EQUIPMENT SUPPORT. CONNECT EXISTING REFRIGERANT PIPING TO CONDENSING UNIT.
2. CONDENSING UNIT MOUNTED ON EQUIPMENT SUPPORT CURB. FLASH AND SEAL WEATHER-TIGHT TO EXISTING ROOF ASSEMBLY.
3. REFRIGERANT PIPING DOWN THROUGH ROOF. PROVIDE PENETRATION CURB. FLASH AND SEAL WEATHER-TIGHT TO EXISTING ROOF ASSEMBLY.
4. REFRIGERANT PIPING DOWN THROUGH ROOF. PROVIDE PENETRATION CURB. FLASH AND SEAL WEATHER-TIGHT TO EXISTING ROOF ASSEMBLY. UTILIZE EXISTING PENETRATIONS.

GANNETT FLEMING

Gannett Fleming, Inc.

1010 Adams Avenue
Valley Forge • PA • 19403
t 610 • 650 • 8156

www.**gannettfleming**.com

READING FIRE DEPARTMENT, ENGINE 7

950 MCKNIGHT ST, READING, PA 19601

HVAC REPLACEMENT / IAQ UPGRADE

NO.	DATE	DESCRIPTION

REVISIONS

BID DOCUMENTS

PROJECT NUMBER: 066446.007

DESIGNED:	MODELED:	CHECKED:
AMF	AMF	MJM

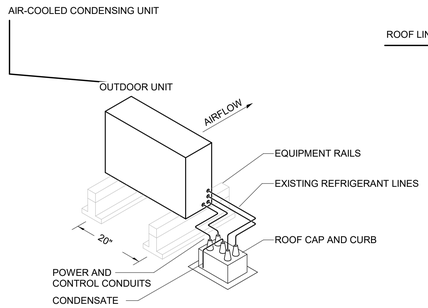
DATE: 22 JULY 2022

DRAWING TITLE: MECHANICAL NEW WORK ROOF PLAN

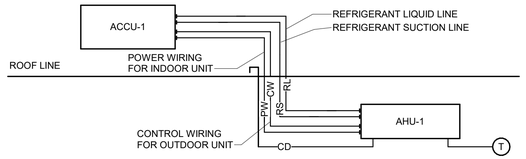
DRAWING NO.: SHEET 7 OF 17

M104

Local File:BIM 360:///066446-City_of_Reading/066446_007_MEP_Central_R21.rvt
Project Number: 066446.007
7/22/2022 7:17:15 PM

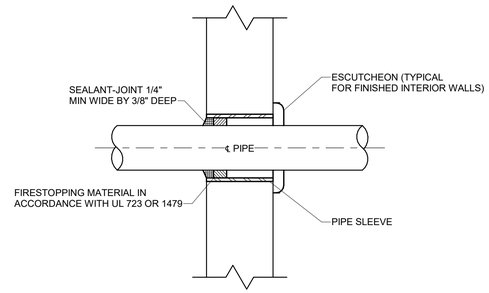


1 OUTDOOR CONDENSING UNIT

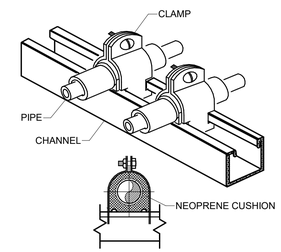


NOTES:
1. INSTALL COPPER TYPE 'ACR' REFRIGERANT PIPING AND ACCESSORIES AS RECOMMENDED BY AND IN ACCORDANCE WITH MFR. AND RECOGNIZED PRACTICES.
2. PIPE SIZES AND CONFIGURATIONS SHALL BE PER MANUFACTURER'S SPECIFICATIONS AND RECOMMENDATIONS.
3. INSULATE BOTH REFRIGERANT LINES PER MFR RECOMMENDATIONS. MIN 1/2".
4. ROUTE OUTDOOR CONTROL WIRING WITH, AND SECURE TO, GAS LINE BETWEEN UNITS.
5. INSTALL REFRIGERANT PIPING PARALLEL OR PERPENDICULAR TO THE LINES OF THE BUILDING STRUCTURE ALLOWING A MINIMUM OF FOUR INCHES CLEARANCE TO EXISTING PIPING, EQUIPMENT, AND ELECTRICAL/CONTROL WIRING.

2 SPLIT SYSTEM PIPING DIAGRAM

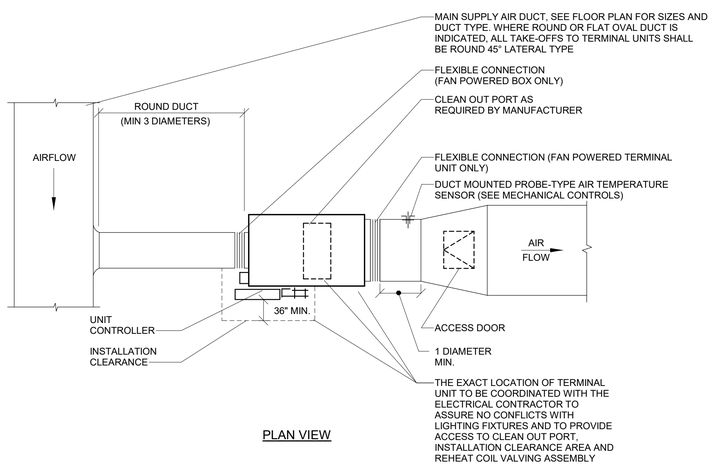


3 PIPING WALL PENETRATION



NOTES:
1. USE GRINNELL PS SERIES POWER-STRUT CHANNEL WITH PS1400 PIPE AND CONDUIT CLAMPS OR EQUIVALENT. CLAMPS TO HAVE CAPTIVE MACHINE SCREW/NYLON LOCKNUT AND A THERMOPLASTIC RUBBER INSERT.
2. REFRIGERANT PIPING TO BE SECURED AS NECESSARY TO PREVENT MOVEMENT, VIBRATION, AND NOISE. MAXIMUM DISTANCE BETWEEN ANCHORAGE POINTS = 6'-0".
3. USE GRINNELL SERIES PS 684 BEAM CLAMPS OR EQUIVALENT TO ATTACH CHANNELS TO ROOF/FLOOR SUPPORT BEAMS.
4. USE EXPANSION TYPE FASTENERS APPROVED FOR USE IN BASE MATERIAL FOR ATTACHING CHANNELS TO INTERIOR WALLS.

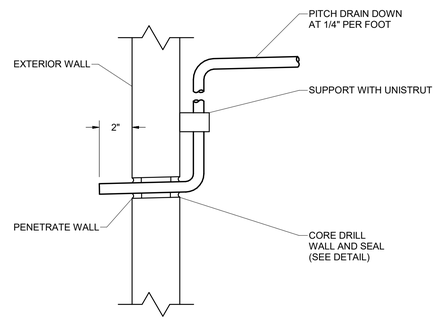
4 REFRIGERANT AND CONDENSATE PIPING SUPPORT



PLAN VIEW

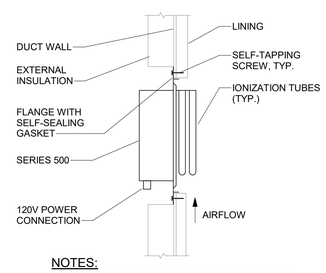
NOTES
1. THE VAV BOX ARRANGEMENT SHALL BE EASILY ACCESSIBLE FROM A SINGLE POINT OF CEILING ACCESS. THIS INCLUDES ACCESS TO THE DAMPER, DAMPER LINKAGE, DAMPER MOTOR, CONTROL ENCLOSURE, DISCONNECT SWITCH. IN ROOMS WHERE A SUPPLY & RETURN BOX ARE LOCATED ADJACENT TO ONE ANOTHER, ACCESS TO ALL COMPONENTS OF BOTH BOXES SHALL BE EASILY ACCESSIBLE FROM A SINLE POINT OF CEILING ACCESS.

5 VAV TERMINAL UNIT INSTALLATION



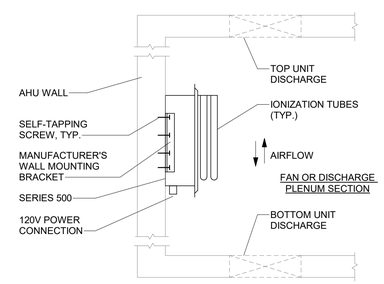
NOTES:
1. PAINT EXPOSED DRAIN TO MATCH ADJACENT WALL.
2. INSTALL TUBING CONTINUOUSLY SLOPING IN DIRECTION OF FLOW WITHOUT LOW SPOTS OR TRAPPED WATER.

6 UNIT VENTILATOR DRAIN PIPING



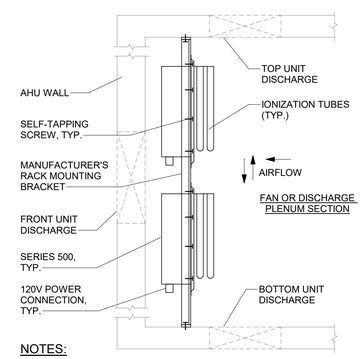
NOTES:
1. INSTALL PER MANUFACTURER'S INSTRUCTIONS.

SERIES 500



NOTES:
1. INSTALL PER MANUFACTURER'S INSTRUCTIONS.
2. ALL ASSEMBLY AND FIXATING HARDWARE SHALL BE AIRCRAFT QUALITY 304 STAINLESS STEEL WITH NYLOCK RETENTION.

WALL MOUNT



NOTES:
1. INSTALL PER MANUFACTURER'S INSTRUCTIONS.
2. ALL ASSEMBLY AND FIXATING HARDWARE SHALL BE AIRCRAFT QUALITY 304 STAINLESS STEEL WITH NYLOCK RETENTION.

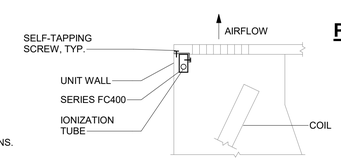
RACK MOUNT

PLENUM/AHU INSTALLATION

NOTES:
1. INSTALL PER MANUFACTURER'S INSTRUCTIONS.

SERIES FC400/MATTERHORN

DUCTWORK INSTALLATION



SERIES FC400

UV/FCU INSTALLATION

7 BIPOLAR IONIZATION INSTALLATION

GANNETT FLEMING

Gannett Fleming, Inc.

1010 Adams Avenue
Valley Forge • PA • 19403
t 610 • 650 • 8156

www.gannettfleming.com

READING FIRE DEPARTMENT, ENGINE 7

950 MCKNIGHT ST, READING, PA 19601

HVAC REPLACEMENT / IAQ UPGRADE

NO.	DATE	DESCRIPTION

REVISIONS

BID DOCUMENTS

PROJECT NUMBER: 066446.007

DESIGNED:	MODELED:	CHECKED:
SAJ	AMF	MJM

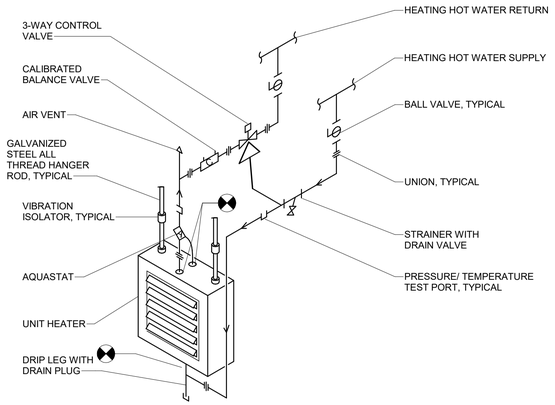
DATE: 22 JULY 2022

DRAWING TITLE: MECHANICAL DETAILS I

DRAWING NO.: M501 SHEET 8 OF 17

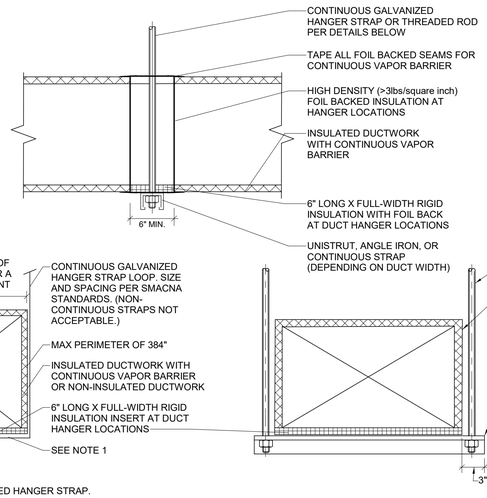
Local File: BIM 360://066446-City_of_Reading/066446_007_MEP_Central_R21.rvt

Project Number: 066446.007
7/22/2022 7:17:16 PM



1 HORIZONTAL HYDRONIC UNIT HEATER

SCALE: NTS

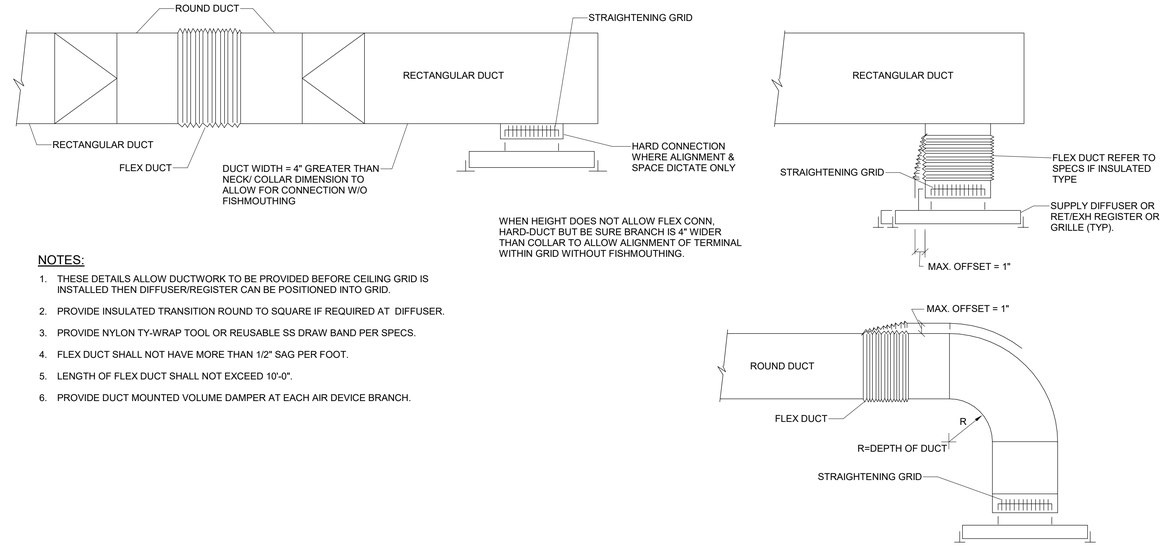


NOTES:

1. CONTINUOUS GALVANIZED HANGER STRAP.
2. FOR CROSS SECTIONAL AREA OF MORE THAN 8 SQ FT, DUCT SHALL BE REINFORCED BY ANGLES ON ALL FOUR SIDES, COORDINATE HANGER AND REINFORCING LOCATIONS.
3. SUPPORTS AND DUCT REINFORCEMENT SHALL BE SPACED AND SIZED AS PER SMACNA STANDARDS.
4. TRAPEZE HANGERS MUST BE USED FOR ALL DUCT OVER 384" PERIMETER.
5. SUPPORTS AND DUCT REINFORCEMENT SHALL BE SPACED AND SIZED AS PER SMACNA STANDARDS. CONTRACTOR SHALL DETAIL AND CONSTRUCT HANGER WITH ZERO DUCT PENETRATIONS.

2 RECTANGULAR DUCT HANGER SUPPORT WITH INSULATION (NO SEISMIC RESTRAINTS)

SCALE: NTS

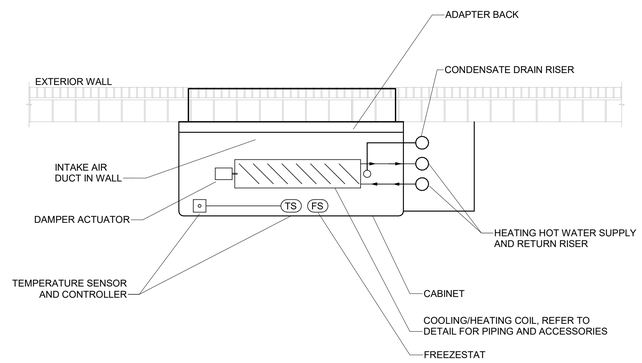


NOTES:

1. THESE DETAILS ALLOW DUCTWORK TO BE PROVIDED BEFORE CEILING GRID IS INSTALLED THEN DIFFUSER/REGISTER CAN BE POSITIONED INTO GRID.
2. PROVIDE INSULATED TRANSITION ROUND TO SQUARE IF REQUIRED AT DIFFUSER.
3. PROVIDE NYLON TY-WRAP TOOL OR REUSABLE SS DRAW BAND PER SPECS.
4. FLEX DUCT SHALL NOT HAVE MORE THAN 1/2" SAG PER FOOT.
5. LENGTH OF FLEX DUCT SHALL NOT EXCEED 10'-0".
6. PROVIDE DUCT MOUNTED VOLUME DAMPER AT EACH AIR DEVICE BRANCH.

3 TYPICAL DIFFUSER AND REGISTER DUCTWORK CONNECTION

SCALE: NTS



4 UNIT VENTILATOR

SCALE: NTS

GANNETT FLEMING

Gannett Fleming, Inc.

1010 Adams Avenue
Valley Forge • PA • 19403
t 610 • 650 • 8156

www.gannettfleming.com

READING FIRE DEPARTMENT, ENGINE 7

950 MCKNIGHT ST, READING, PA 19601

HVAC REPLACEMENT / IAQ UPGRADE

NO.	DATE	DESCRIPTION

REVISIONS

BID DOCUMENTS

PROJECT NUMBER: 066446.007

DESIGNED:	MODELED:	CHECKED:
SAJ	AMF	MJM

DATE: 22 JULY 2022

DRAWING TITLE: MECHANICAL DETAILS II

DRAWING NO.: M502

SHEET 9 OF 17

Local File: BIM 360://066446-City_of_Reading/066446_007_MEP_Central_R21.rvt

Project Number: 066446.007
7/22/2022 7:17:16 PM

AIR HANDLING UNIT SCHEDULE

MARK	SUPPLY AIRFLOW (CFM)	MIN OA AIRFLOW (CFM)	FAN ESP IN W.C.	HOT WATER HEATING PERFORMANCE							DX COIL COOLING PERFORMANCE						CONTROL VALVE TYPE	ELECTRICAL DATA					BASIS OF DESIGN		WEIGHT (LB)	REMARKS
				MBH	EWT (F°)	LWT (F°)	GPM	EAT (F°)	LAT (F°)	HEATING WPD (FT/H2O)	SENSIBLE (MBH)	TOTAL (MBH)	EDB (F°)	EWB (F°)	LDB (F°)	LWB (F°)		V	PH	HZ	MCA	FLA (A)	MANUFACTURER	MODEL		
AHU -1	460	230	1.92	26	180	164	2.6	38	90	4.11	16.33	24.69	86	70	53	53	3 - WAY MODULATING	230	1	60	5.3	4.3	TRANE	BCHE018	250	1,2,3,4,5

NOTES:
1. PROVIDE WITH UNITARY VAV CONTROLLER. PROVIDE LOW VOLTAGE FROM THIS UNIT TO BE THE TWO (2) VAVs.
2. PROVIDE WITH INTEGRAL DISCONNECT SWITCH.
3. PROVIDE WITH REMOTE PROGRAMMABLE THERMOSTAT.
4. PROVIDE WITH MERV-13 FILTERS.
5. PROVIDE WITH INTEGRAL BPI UNIT.

HYDRONIC UNIT HEATER SCHEDULE

MARK	TYPE	HEAT OUTPUT (MBH)	WATER FLOW RATE (GPM)	AIRFLOW RATE(CFM)	MAX WPD (FT/H20)	EWT (F)	LWT (F)	ELECTRICAL DATA				MOUNTING HEIGHT (FT./AFF)	APPROX. OPERATING WEIGHT (LBS)	BASIS OF DESIGN		REMARKS
								V	Ø	HZ	HP			MANUFACTURER	MODEL	
UH-101A	HORIZONAL CEILING MOUNTED	61.0	6.1	1400	0.24	180	160	115	1	60	0.17	12'-0"	47	TRANE	SERPENTINE COIL	1, 2
UH-101B	HORIZONAL CEILING MOUNTED	34.8	3.5	750	0.12	180	160	115	1	60	0.17	12'-0"	41	TRANE	SERPENTINE COIL	1, 2
UH-101C	HORIZONAL CEILING MOUNTED	34.8	3.5	750	0.12	180	160	115	1	60	0.17	12'-0"	41	TRANE	SERPENTINE COIL	1, 2
UH-101D	HORIZONAL CEILING MOUNTED	61.0	6.1	1400	0.24	180	160	115	1	60	0.17	12'-0"	47	TRANE	SERPENTINE COIL	1, 2
UH-B01	HORIZONAL CEILING MOUNTED	34.8	3.5	750	0.12	180	160	115	1	60	0.17	12'-0"	41	TRANE	SERPENTINE COIL	1, 2
UH-B02	HORIZONAL CEILING MOUNTED	26.1	2.7	550	0.09	180	160	115	1	60	0.17	12'-0"	36	TRANE	SERPENTINE COIL	1, 2
UH-B03	HORIZONAL CEILING MOUNTED	34.8	3.5	750	0.12	180	160	115	1	60	0.17	12'-0"	41	TRANE	SERPENTINE COIL	1, 2

NOTES:
1. PROVIDE WITH OSHA FAN GUARDS AND ADJUSTABLE AIR DEFLECTION LOUVERS.
2. PROVIDE WITH FACTORY-INSTALLED AND WIRED UNIT-MOUNTED DISCONNECT SWITCH.
3. PROVIDE WITH REMOTE THERMOSTAT WITH ON/OFF/AUTO SWITCH.

UNIT VENTILATOR SCHEDULE

MARK	TYPE	SUPPLY AIRFLOW (CFM)	OUTSIDE AIRFLOW (CFM)	HEATING HOT WATER COIL PERFORMANCE					DX COOLING COIL PERFORMANCE							ELECTRICAL DATA						APPROX. OPERATING WEIGHT (LBS)	BASIS OF DESIGN		REMARKS
				HEAT OUTPUT (MBH)	WATER FLOW (GPM)	EWT (F)	LWT (F)	MAX WPD (FT/H20)	TOTAL CAPACITY (MBH)	SENSIBLE CAPACITY (MBH)	EDB (F)	EWB (F)	LDB (F)	LWB (F)	FLUID TYPE	V	Ø	HZ	HP	MCA	MOCP		MANUFACTURER	MODEL	
UV-112	HORIZONTAL UNIT VENTILATOR	740	90	40.4	4.0	180	160	5.5	24.48	17.33	80	67	56	54	R-410A	115	1	60	0.25	10.2	15.0	130	TRANE	VUVE07500	1, 2, 3, 4, 5
UV-112A	HORIZONTAL UNIT VENTILATOR	590	115	40.4	4.0	180	160	5.5	24.48	17.33	80	67	56	54	R-410A	115	1	60	0.25	10.2	15.0	130	TRANE	VUVE07500	1, 2, 3, 4, 5
UV-113A	HORIZONTAL UNIT VENTILATOR	465	125	40.4	4.0	180	160	5.5	24.48	17.33	80	67	56	54	R-410A	115	1	60	0.25	10.2	15.0	130	TRANE	VUVE07500	1, 2, 3, 4, 5
UV-113B	HORIZONTAL UNIT VENTILATOR	465	125	40.4	4.0	180	160	5.5	24.48	17.33	80	67	56	54	R-410A	115	1	60	0.25	10.2	15.0	130	TRANE	VUVE07500	1, 2, 3, 4, 5

NOTES:
1. PROVIDE WITH UNITARY CONTROLLER AND VARIABLE SPEED SUPPLY FAN.
2. PROVIDE WITH INTEGRAL DISCONNECT SWITCH.
3. PROVIDE WITH REMOTE PROGRAMMABLE THERMOSTAT.
4. PROVIDE WITH MERV-8 FILTERS.
5. PROVIDE WITH INTEGRAL BPI UNIT AND 208/120V TRANSFORMER. BPI UNIT TO BE POWERED FROM SAME CIRCUIT AS UNIT VENTILATOR.

EXHAUST FAN SCHEDULE

MARK	SERVES	TYPE	AIRFLOW (CFM)	ESP (IN W.C.)	FAN SPEED (RPM)	ELECTRICAL				OPERATING WEIGHT (LB)	MANUFACTURER	MODEL	REMARKS
						WATTS	PHASE	VOLTAGE	HERTZ				
EF-103	KITCHEN EXHAUST	WALL-MOUNTED	180	0.10	1160	56.9	1	115	60	12	LOREN COOK	GN-168	1, 2, 3
EF-106	SHOWER EXHAUST	IN-LINE	460	0.25	1500	183	1	115	60	30	LOREN COOK	GC-128	1, 2, 3
EF-115	TOILET EXHAUST	CEILING-MOUNTED	50	0.13	750	29	1	115	60	11	LOREN COOK	GN-168	1, 2, 3

NOTES:
1. EXHAUST FAN SHALL BE UL 705 LISTED AND BEAR THE AMCA CERTIFIED RATINGS SEAL FOR SOUND AND AIR PERFORMANCE.
2. PROVIDE WITH DISCONNECT SWITCH.
3. PROVIDE WITH REMOTE SPEED SELECTOR SWITCH.

AIR COOLED CONDENSING UNIT SCHEDULE

MARK	ASSOCIATED UNIT	COOLING CAPACITY (MBH)	ELECTRICAL DATA					REFRIGERANT TYPE	CIRCUIT QTY.	COMPRESSOR QTY.	OPERATING WEIGHT (LBS)	BASIS OF DESIGN	
			V	Ø	RATED LOAD AMPS	MCA	MOCP					MANUFACTURER	MODEL
ACCU-1	AHU-1	30,000	208	1	12.8	17	25	R-410A	1	1	184	TRANE	4TTR6030J
ACCU-112	UV-112	30,000	208	1	12.8	17	25	R-410A	1	1	184	TRANE	4TTR6030J
ACCU-112A	UV-112A	30,000	208	1	12.8	17	25	R-410A	1	1	184	TRANE	4TTR6030J
ACCU-113A	UV-113A	30,000	208	1	12.8	17	25	R-410A	1	1	184	TRANE	4TTR6030J
ACCU-113B	UV-113B	30,000	208	1	12.8	17	25	R-410A	1	1	184	TRANE	4TTR6030J

VARIABLE AIR VOLUME (VAV) TERMINAL UNIT SCHEDULE

MARK	SERVES	VAV BOX TYPE	INLET SIZE, (IN)	MAX FLOW (CFM)	ELECTRICAL DATA			OPERATING WEIGHT (LBS)	BASIS OF DESIGN		REMARKS
					V	PH	HZ		MANUFACTURER	MODEL	
VAV-1	KITCHEN	RECTANGULAR IN/OUT; DAMPER ONLY	10" W x 8" H	385	24	1	60	21	ZONEX	STCD	1
VAV-2	WOMEN'S DORMITORY	ROUND IN/OUT; DAMPER ONLY	6" DIAMETER	75	24	1	60	24	ZONEX	STMPD	1

NOTES:
1. SET DAMPERS TO BE 70% OPEN WHEN FULLY CLOSED.

CONVECTOR HEATER SCHEDULE

MARK	HEATING CAPACITY (MBH)	HEATING FLUID FLOW (GPM)	HEATING WPD (FT/H2O)	EWT (°F)	LWT (°F)	HEATER LENGTH (FT)	MANUFACTURER	MODEL	REMARKS
CVC-102	5	0.15	0.044	180	160	48	TRANE	FWG-A-424	1
CVC-105	5	0.15	0.044	180	160	48	TRANE	FWG-A-424	1
CVC-107	5	0.15	0.044	180	160	48	TRANE	FWG-A-424	1
CVC-108	2.5	0.15	0.044	180	160	24	TRANE	FWG-A-424	1
CVC-109	7	0.15	0.044	180	160	64	TRANE	FWG-A-424	1
CVC-117	5	0.15	0.044	180	160	48	TRANE	FWG-A-424	1
CVC-118	5	0.15	0.044	180	160	48	TRANE	FWG-A-424	1

NOTES:
1. PROVIDE WITH REMOTE THERMOSTAT WITH ON/OFF/AUTO SWITCH.

BIPOLAR IONIZATION SCHEDULE

MARK	QTY	AIRFLOW (CFM)	ELECTRICAL DATA		IONS/CM3	BASIS OF DESIGN		DESCRIPTION	REMARKS
			V	Ø		MANUFACTURER	MODEL		
BPI-101A	1	200	120	1	1500	ATMOSAIR	EVEREST	STANDALONE EQUIPMENT	1
BPI-101B	1	200	120	1	1500	ATMOSAIR	EVEREST	STANDALONE EQUIPMENT	1
BPI-101C	1	200	120	1	1500	ATMOSAIR	EVEREST	STANDALONE EQUIPMENT	1
BPI-101D	1	200	120	1	1500	ATMOSAIR	EVEREST	STANDALONE EQUIPMENT	1
BPI-103	1	430	120	1	1500	ATMOSAIR	MATTERHORN 1000	D TUBE SIZE, DUCT-MOUNTED	1
BPI-109	1	100	120	1	1500	ATMOSAIR	EVEREST	STANDALONE EQUIPMENT	1
BPI-112	1	620	120	1	1500	ATMOSAIR	FC-400	MOUNTED IN UV	1
BPI-112A	1	490	120	1	1500	ATMOSAIR	FC-400	MOUNTED IN UV	1
BPI-113A	1	330	120	1	1500	ATMOSAIR	FC-400	MOUNTED IN UV	1
BPI-113B	1	330	120	1	1500	ATMOSAIR	FC-400	MOUNTED IN UV	1
BPI-B01	1	100	120	1	1500	ATMOSAIR	EVEREST	STANDALONE EQUIPMENT	1
BPI-B03	1	100	120	1	1500	ATMOSAIR	EVEREST	STANDALONE EQUIPMENT	1

NOTES:
1. PROVIDE WITH 1 SET OF REPLACEMENT TUBES.

GANNETT FLEMING

Gannett Fleming, Inc.

1010 Adams Avenue
Valley Forge • PA • 19403
t 610 • 650 • 8156

www.gannettfleming.com

READING FIRE DEPARTMENT, ENGINE 7

950 MCKNIGHT ST, READING, PA 19601

HVAC REPLACEMENT / IAQ UPGRADE

NO.	DATE	DESCRIPTION

REVISIONS

BID DOCUMENTS

PROJECT NUMBER: 066446.007

DESIGNED: SAJ | MODELED: AMF | CHECKED: MJM

DATE: 22 JULY 2022

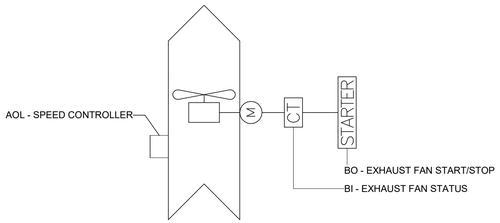
DRAWING TITLE: MECHANICAL SCHEDULES

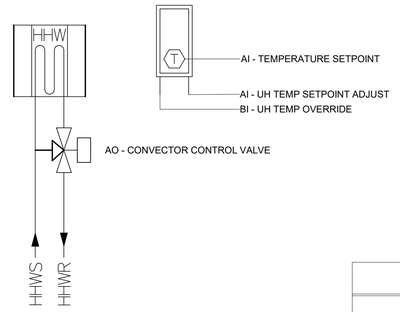
DRAWING NO.: M601

SHEET 10 OF 17

Local File:BIM 360://066446-City_of_Reading/066446_007_MEP_Central_R21.rvt

Project Number: 066446.007
7/22/2022 7:17:16 PM





EXHAUST FAN

A. GENERAL
   1. THE EXHAUST FAN SHALL BE CONTROLLED VIA A POSITION SWITCH IN THE OCCUPIED SPACE.
B. MOTOR CONTROL, START-UP AND SHUTDOWN
   1. THE OCCUPIED AREA SWITCH WILL COMPLETE A 120V CIRCUIT ENABLING THE EXHAUST FAN TO START AND THE MANUAL SPEED CONTROLLER SWITCH SHALL MODULATE RPM OF FAN.
C. STAGING - NOT APPLICABLE
D. TEMPERATURE CONTROL – NOT APPLICABLE
E. HUMIDITY CONTROL – NOT APPLICABLE
F. STATIC PRESSURE AND FLOW CONTROL – NOT APPLICABLE
G. STAND-BY POWER MODE – NOT APPLICABLE
H. SAFETIES – NOT APPLICABLE
I. FAILURES - NOT APPLICABLE
J. FIRE ALARM MODE - NOT APPLICABLE

EXHAUST FAN

POINT NAME	AI	AO	BI	BO
EXHAUST FAN START/STOP				X
EXHAUST FAN STATUS			X	
SPEED CONTROLLER		X		

CONVECTOR

A. GENERAL
   1. THE HEATER SHALL READ LOCAL SPACE TEMPERATURE AND MODULATE THE HEATING HOT WATER VALVE TO MEET THE SPACE TEMPERATURE SETON (60 DEGREES-FA, ADJUSTABLE).
B. MOTOR CONTROL, START-UP AND SHUTDOWN - NOT APPLICABLE
C. STAGING – NOT APPLICABLE
D. TEMPERATURE CONTROL
   1. HEATING: THE CONTROLLER SHALL MEASURE THE ZONE TEMPERATURE AND MODULATE THE HEATING TO MAINTAIN ITS HEATING SETON. THE HEATING SHALL BE ENABLED WHENEVER:
      a. THE ZONE TEMPERATURE IS BELOW HEATING SETON (60 DEGREES-FA, ADJUSTABLE).
E. HUMIDITY CONTROL – NOT APPLICABLE
F. STATIC PRESSURE AND FLOW CONTROL
   1. THE HEATING HOT WATER CONTROL VALVE SHALL MODULATE TO MAINTAIN SPACE SETON.
G. STAND-BY POWER MODE – NOT APPLICABLE
H. SAFETIES – NOT APPLICABLE
I. FAILURES – NOT APPLICABLE
J. FIRE ALARM MODE - NOT APPLICABLE

CONVECTOR

POINT NAME	AI	AO	BO	BI
CONTROL VALVE POSITION		X		
TEMPERURE SENSOR	X			
TEMPERATURE SETPOINT	X			

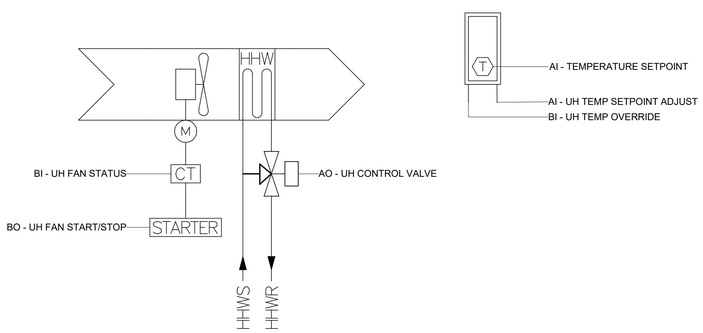
CONTROL WIRING RACEWAY SCHEDULE

LOCATION	RACEWAY TYPE
EXPOSED IN CLASSROOMS, OFFICES, CORRIDORS, ETC.	ELECTRICAL METALLIC TUBING
CONCEALED ABOVE DROP CEILINGS AND WITHIN WALLS	ELECTRICAL METALLIC TUBING
EXPOSED IN MECHANICAL ROOMS AND ELECTRICAL ROOMS	RIGID METAL CONDUIT
OUTDOORS AND WET AREAS	RIGID METAL CONDUIT

ALL CONTROL WIRING SHALL BE PROVIDED IN CONDUIT. WIREMOLD IS NOT ACCEPTABLE.

CONTROL HARDWARE & POINTS, ABBREVIATIONS AND SYMBOLS

DJ	ADJUSTABLE
AI	ANALOG INPUT
AOL	ANALOG OUTPUT
AUX	AUXILIARY
AV	ANALOG VALUE
BI	BINARY INPUT
BO	BINARY OUTPUT
BCU	BUILDING CONTROL UNIT
BAS	BUILDING AUTOMATION SYSTEM
BV	BINARY VALUE
CO2	CARBON DIOXIDE
COM	CARBON MONOXIDE MONITOR
DDCP	DIRECT DIGITAL CONTROL PANEL
DBT	DRY BULB TEMPERATURE
WBT	WET BULB TEMPERATURE
DPT	DEW POINT TEMPERATURE
EAT	ENTERING AIR TEMPERATURE
LAT	LEAVING AIR TEMPERATURE
EAH	ENTERING AIR MOISTURE
LAH	LEAVING AIR MOISTURE
(E)	EXISTING
HO/S	REMOTE HORN STROBE UNIT FOR CARBON MONOXIDE ALARM
HI	HUMAN MACHINE INTERFACE
RH	RELATIVE HUMIDITY
NC	NORMALLY CLOSED
NO	NORMALLY OPEN
OEM	ORIGINAL EQUIPMENT MANUFACTURER
TEMP	TEMPERATURE



UNIT HEATER

A. GENERAL
   1. THE HEATER SHALL READ LOCAL SPACE TEMPERATURE AND CYCLE THE SUPPLY FAN AND MODULATE THE HEATING HOT WATER VALVE TO MEET THE SPACE TEMPERATURE SETON (60 DEGREES-FA, ADJUSTABLE).
B. MOTOR CONTROL, START-UP AND SHUTDOWN
   1. UPON A READING 1F BELOW SPACE SETON THE UNIT HEATER SHALL ENGAGE TO MAINTAIN SPACE TEMPERATURE. FIRST THE SUPPLY FAN SHALL TURN ON AND PROVE ON FOLLOWED BY THE HEATING HOT WATER CONTROL VALVE MODULATING.
C. STAGING – NOT APPLICABLE
D. TEMPERATURE CONTROL
   1. HEATING: THE CONTROLLER SHALL MEASURE THE ZONE TEMPERATURE AND MODULATE THE HEATING.WHITEWATER VALVE TO MAINTAIN ITS HEATING SETON. THE HEATING SHALL BE ENABLED WHENEVER:
      a. THE ZONE TEMPERATURE IS BELOW HEATING SETON (60 DEGREES-FA, ADJUSTABLE).
      b. AND THE SUPPLY FAN STATUS IS ON.
E. HUMIDITY CONTROL – NOT APPLICABLE
F. STATIC PRESSURE AND FLOW CONTROL
   1. THE HEATING HOT WATER CONTROL VALVE SHALL MODULATE TO MAINTAIN SPACE SETPOINT.
G. STAND-BY POWER MODE – NOT APPLICABLE
H. SAFETIES – NOT APPLICABLE
I. FAILURES – NOT APPLICABLE
J. FIRE ALARM MODE
   1. THE HEATER SHALL CONTINUE TO OPERATE UPON A FIRE ALARM SIGNAL.

UNIT HEATER

POINT NAME	AI	AO	BI	BO
UH CONTROL VALVE		X		
UH TEMPERATURE	X			
UH TEMP OVERRIDE			X	
UH TEMP SETPOINT ADJUST	X			
UH TEMPERATURE	X			
UH TEMP OVERRIDE			X	
UH TEMP SETPOINT ADJUST	X			
UH FAN START/STOP				X
UH FAN STATUS			X	

GENERAL BUILDING CONTROL NOTES

1. ALL EQUIPMENT TO BE CONTROLLED BY LINE VOLTAGE THERMOSTAT OR PROGRAMMABLE THERMOSTAT AS NOTED ON THE PLANS.

CONTROL SHEET NOTES

1. ADDITIONAL CONTROLLED MECH ITEMS (EFs, PUMPS, DAMPERS... ETC) SHALL BE CONTROLLED BY DISTRIBUTED NACs. SEE FLOOR PLANS FOR NAC LOCATIONS. REFER TO CONTROL DIAGRAMS FOR MORE.

GANNETT FLEMING

Gannett Fleming, Inc.

1010 Adams Avenue
Valley Forge • PA • 19403
t 610 • 650 • 8156

www.gannettfleming.com

READING FIRE DEPARTMENT, ENGINE 7

950 MCKNIGHT ST, READING, PA 19601

HVAC REPLACEMENT / IAQ UPGRADE

NO.	DATE	DESCRIPTION

REVISIONS

BID DOCUMENTS

PROJECT NUMBER: 066446.007

DESIGNED:	MODELED:	CHECKED:
SAJ	AMF	MJM

DATE: 22 JULY 2022

DRAWING TITLE: MECHANICAL

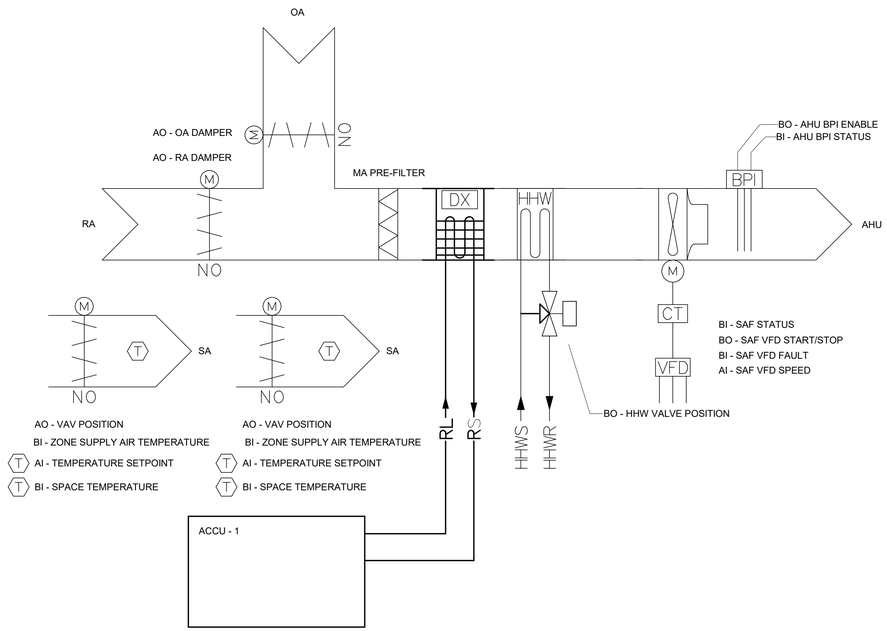
MECHANICAL CONTROL DIAGRAMS I

DRAWING NO.: SHEET 11 OF 17

M801

Local File:BIM 360://066446-City_of_Reading/066446_007_MEP_Central_R21.rvt

Project Number: 066446.007
7/22/2022 7:17:17 PM



AIR HANDLING UNIT (AHU-1, 2, 3) SEQUENCE OF OPERATION

A. GENERAL
   1. SEE SYSTEM DIAGRAM FOR SYSTEM COMPONENT OF THE AIR HANDLING UNIT.
   2. THE EQUIPMENT SHALL BE COMPLETELY CONTROLLED BY THE ON-BOARD UNITARY CONTROLLER (UC) . THE SYSTEM SHALL BE AVAILABLE 24 HOURS PER DAY, 365 DAYS PER YEAR UNLESS OTHERWISE SHUTDOWN ON SAFETIES.
   3. ALL CONTROLS SHALL BE BY DIRECT DIGITAL CONTROLS (DDC). ALL SETPOINTS LISTED WITHIN THE SEQUENCE SHALL BE ADJUSTABLE BY USER.
   4. CONSTANT VOLUME AIR HANDLING UNIT SHALL BE CONTROLLED BY A CONSTANT AIR TEMPERATURE DEPENDING ON WHICH MORE SYSTEM IS IN SEE TEMPERATURE CONTROL BELOW.

B. MOTOR CONTROL, START-UP AND SHUTDOWN
   1. WHEN THE SYSTEM IS COMMANDED TO START THE SUPPLY FAN SHALL START AND SYSTEM SHALL RESPOND TO TEMPERATURE CONTROL BELOW.

C. STAGING – NOT APPLICABLE

D. TEMPERATURE CONTROL
   1. THE UNIT SHALL RUN ACCORDING TO A USER DEFINABLE TIME SCHEDULE IN THE FOLLOWING MODES:
      a. THE UNIT SHALL MAINTAIN A CONSTANT DISCHARGE TEMPERATURE
         1. A 55°F (ADJ.) COOLING DISCHARGE SETPOINT.
         2. A 85°F (ADJ.) HEATING DISCHARGE SETPOINT.

E. HUMIDITY CONTROL – NOT APPLICABLE

F. STATIC PRESSURE AND FLOW CONTROL

G. STAND-BY POWER MODE – NOT APPLICABLE

H. SAFETIES
   1. THE UNIT SHALL OPERATE TO MEET ITS FACTORY SAFETIES.

I. FAILURES
   1. ANY FACTORY DEFINED FAILURES SHALL BE DISPLAYED ON THE PROGRAMMABLE THERMOSTAT.

J. FIRE ALARM MODE – NOT APPLICABLE

GANNETT FLEMING

Gannett Fleming, Inc.

1010 Adams Avenue
Valley Forge • PA • 19403
t 610 • 650 • 8156

www.gannettfleming.com

READING FIRE DEPARTMENT, ENGINE 7

950 MCKNIGHT ST, READING, PA 19601

HVAC REPLACEMENT / IAQ UPGRADE

NO.	DATE	DESCRIPTION

REVISIONS

BID DOCUMENTS

PROJECT NUMBER: 066446.007

DESIGNED:	MODELED:	CHECKED:
SAJ	AMF	MJM

DATE: 22 JULY 2022

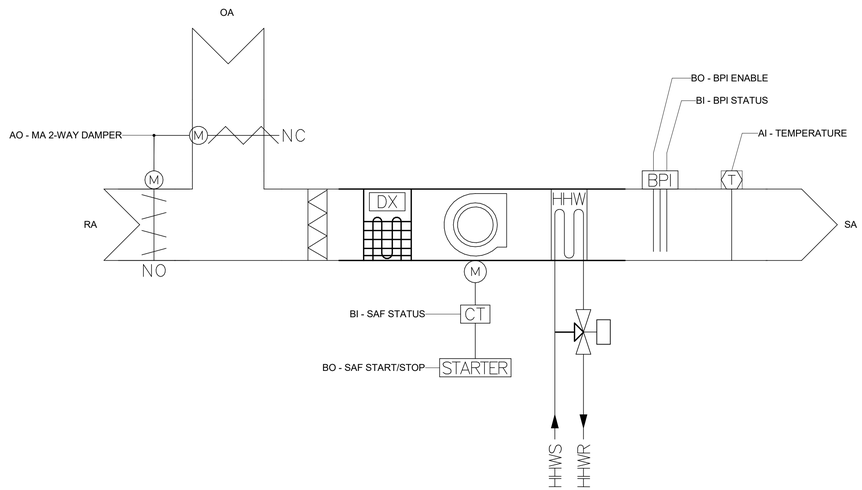
DRAWING TITLE:
MECHANICAL

MECHANICAL CONTROL DIAGRAMS II

DRAWING NO.: SHEET 12 OF 17

M802

Local File:BIM 360://066446-City_of_Reading/066446_007_MEP_Central_R21.rvt
Project Number: 066446.007
7/22/2022 7:17:17 PM



UNIT VENTILATOR SEQUENCE OF OPERATION

A. GENERAL
   1. THE UNIT VENTILATOR CONSISTS OF A PACKAGED SYSTEM WITH DUAL TEMPERATURE COIL FOR HEATING/COOLING RESPECTIVELY AND BYPASS DAMPER FOR TEMPERATURE CONTROL.
   2. THE EQUIPMENT SHALL BE COMPLETELY CONTROLLED BY THE ON-BOARD UNITARY CONTROLLER (UC). THE SYSTEM SHALL BE AVAILABLE 24 HOURS PER DAY, 365 DAYS PER YEAR UNLESS OTHERWISE SHUTDOWN ON SAFETIES.
   3. ALL CONTROLS SHALL BE BY DIRECT DIGITAL CONTROLS (DDC). ALL SETPOINTS LISTED WITHIN THE SEQUENCE SHALL BE ADJUSTABLE BY USER.

B. MOTOR CONTROL START-UP AND SHUTDOWN
   1. WHEN THE SYSTEM IS COMMANDED TO START THE OUTDOOR AIR DAMPER SHALL BE SET TO MINIMUM POSITIONS (ONLY IN OCCUPIED MODE; IN UNOCCUPIED MODE THE SYSTEM SHALL OPERATE IN RECIRCULATION MODE ONLY WITH THE OUTSIDE AIR DAMPER CLOSED).
   2. THE RETURN AIR DAMPER SHALL BE FULLY OPEN AND THE SUPPLY FAN SHALL START.
   3. UPON A CALL FOR SHUTDOWN, THE SUPPLY FAN SHALL SHUTDOWN AND OUTSIDE AIR DAMPER SHALL CLOSE.

C. STAGING – NOT APPLICABLE

D. TEMPERATURE CONTROL
   1. THE UNIT SHALL RUN ACCORDING TO A USER DEFINABLE TIME SCHEDULE FOR OCCUPIED AND UNOCCUPIED MODES OF OPERATION VIA PROGRAMMABLE THERMOSTAT ZONE SETPOINT ADJUST:
      a. THE OCCUPANT SHALL BE ABLE TO ADJUST THE ZONE TEMPERATURE HEATING AND COOLING SETPOINTS AT THE ZONE SENSOR.
   2. ZONE UNOCCUPIED OVERRIDE:
      a. A TIMED LOCAL OVERRIDE CONTROL SHALL ALLOW AN OCCUPANT TO OVERRIDE THE SCHEDULE AND PLACE THE UNIT INTO AN OCCUPIED MODE FOR AN ADJUSTABLE PERIOD OF TIME. AT THE EXPIRATION OF THIS TIME, CONTROL OF THE UNIT SHALL AUTOMATICALLY RETURN TO THE SCHEDULE. THE COOLING MODE SHALL BE ENABLED WHENEVER:
   3. COOLING MODE: THE CONTROLLER SHALL ENABLE THE DX COOLING COIL AND MODULATE THE DX COIL CAPACITY TO MAINTAIN A SPACE TEMPERATURE SETPOINT.
   4. HEATING MODE: THE UNITARY CONTROLLER SHALL MODULATE THE HOT WATER CONTROL VALVE TO MAINTAIN THE HEATING SPACE TEMPERATURE SETPOINT.

E. HUMIDITY CONTROL – NOT APPLICABLE

F. STATIC PRESSURE AND FLOW CONTROL – NOT APPLICABLE

G. STAND-BY POWER MODE – NOT APPLICABLE

H. SAFETIES
   1. THE UNIT SHALL OPERATE TO MEET ITS FACTORY SAFETIES.

I. FAILURES
   1. ANY FACTORY DEFINED FAILURES SHALL BE DISPLAYED ON THE PROGRAMMABLE THERMOSTAT.

I. FIRE ALARM MODE – NOT APPLICABLE

GANNETT FLEMING

Gannett Fleming, Inc.

1010 Adams Avenue
Valley Forge • PA • 19403
t 610 • 650 • 8156

www.**gannettfleming**.com

READING FIRE DEPARTMENT, ENGINE 7

950 MCKNIGHT ST, READING, PA 19601

HVAC REPLACEMENT / IAQ UPGRADE

NO.	DATE	DESCRIPTION

REVISIONS

BID DOCUMENTS

PROJECT NUMBER: 066446.007

DESIGNED:	MODELED:	CHECKED:
SAJ	AMF	MJM

DATE: 22 JULY 2022

DRAWING TITLE: MECHANICAL CONTROL DIAGRAMS III

DRAWING NO.: SHEET 13 OF 17

M803

Local File:BIM 360://066446-City_of_Reading/066446_007_MEP_Central_R21.rvt
Project Number: 066446.007
7/22/2022 7:17:17 PM

ABBREVIATIONS

A or AMP	AMPERE
AC	ALTERNATING CURRENT
AFF	ABOVE FINISH FLOOR
AFG	ABOVE FINISH GRADE
AIC	AMPERE INTERRUPTING CAPACITY
AS	AMMETER SELECTOR SWITCH AMMETER SELECTOR SWITCH
ATS	AUTOMATIC TRANSFER SWITCH
AUTO	AUTOMATIC
AWG	AMERICAN WIRE GAUGE
BLDG	BUILDING
C	CONDUIT
CP	CONTROL PANEL
CPT	CONTROL POWER TRANSFORMER
DISC	DISCONNECT
DIV	DIVISION
DS	DISCONNECT SWITCH
EC	ELECTRICAL CONTRACTOR
ECH	ELECTRIC CABINET HEATER
EF	EXHAUST FAN
EMT	ELECTRICAL METALLIC TUBING(CONDUIT) ELECTRICAL METALLIC TUBING(CONDUIT)
EP	EXPLOSION PROOF
EUH	ELECTRIC UNIT HEATER
F/T	FEED THROUGH
FAAP	FIRE ALARM ANNUNCIATOR PANEL
FACP	FIRE ALARM CONTROL PANEL
FRE	FIBERGLASS REINFORCED EPOXY
FU	FUSE
GFI	GROUND FAULT INTERRUPTER
GRD	GROUND
HID	HIGH INTENSITY DISCHARGE
HP	HORSEPOWER
HPS	HIGH PRESSURE SODIUM
HVAC	HEATING - VENTILATION - AIR CONDITIONING
IG	ISOLATED GROUND
IND	INDUSTRIAL
JIC	JOINT INDUSTRIAL COUNCIL
KV	KILOVOLT
KVA	KILOVOLT AMPERE
KW	KILOWATT
LA	LIGHTNING ARRESTOR
LC	LIGHTING CONTACTOR
LTG	LIGHTING
MCC	MOTOR CONTROL CENTER
MLO	MAIN LUG ONLY
MOA	MULTI OUTLET ASSEMBLY
MOD	MOTOR OPERATED DAMPER
MS	MOTOR STARTER
MTD	MOUNTED
MV	MEDIUM VOLTAGE
N/A	NOT APPLICABLE
NC	NORMALLY CLOSED
NO	NORMALLY OPEN
No	NUMBER
PMT	PAD MOUNTED TRANSFORMER
PNL	PANEL
PT	POTENTIAL TRANSFORMER
PVC	POLYVINYL CHLORIDE (CONDUIT)
RECP	RECEPTACLE
RGS	RIGID GALVANIZED STEEL(CONDUIT)
RVAT	REDUCED VOLTAGE AUTOTRANSFORMER
RVSS	REDUCED VOLTAGE SOLID STATE
SC	SURGE CAPACITOR
SPD	SURGE PROTECTION DEVICE
SW	SWITCH
SWBD	SWITCHBOARD
TC	CABLE TRAY - CABLE
TTB	TELEPHONE TERMINAL BOARD
TTC	TELEPHONE TERMINAL CABINET
TYP	TYPICAL
UH	UNIT HEATER
UL	UNDERWRITER LABORATORIES
UON	UNLESS OTHERWISE NOTED
UPS	UNINTERRUPTIBLE POWER SUPPLY
V	VOLT
VS	VOLTMETER SELECTOR SWITCH
W	WIRE
WP	WEATHERPROOF
XFMR	TRANSFORMER
1-PH	SINGLE PHASE
3-PH	THREE PHASE

GENERAL POWER

C	DENOTES MOUNTED 6" ABOVE COUNTER TOP
	SINGLE RECEPTACLE
	DUPLEX RECEPTACLE, NEMA 5-20R
GF	RECEPTACLE WIRED TO GROUND FAULT CIRCUIT BREAKER
GFCI	GROUND FAULT CIRCUIT INTERRUPTER
S - SWITCHED	ISOLATED GROUND
S	SWITCHED
TR	TAMPER RESISTANT
WP	WEATHER RESISTANT RECEPTACLE W/IN-USE COVER (WET LOCATION)
WR	WEATHER RESISTANT RECEPTACLE W/NON-IN-USE COVER (DAMP LOCATION)
TV	DUPLEX RECEPTACLE, NEMA 5-20R RECEPTACLE FOR TV MONITOR
	DOUBLE DUPLEX RECEPTACLE
	DUPLEX RECEPTACLE DEDICATED FOR UPS POWER
	SPECIAL PURPOSE RECEPTACLE (AMPACITY AS NOTED) A - 30A, 2 POLE + GROUND, 208VAC, 1-PHASE, NEMA 4X, IP66/IP67 B - 30A, 1POLE + NEUTRAL + GROUND, 125VAC, NEMA 4X, IP66/IP67 C - 30A, 3 POLE + GROUND, 208VAC, 3-PHASE, NEMA 4X, IP66/IP67 D - 60A, 3 POLE + GROUND, 208VAC, 3-PHASE, NEMA 4X, IP66/IP67 E - 30A, 2 POLE + GROUND, 250VAC, 1-PHASE, TWISTLOCK F - 30A, 1 POLE + NEUTRAL + GROUND, 125VAC, TWISTLOCK
	DUPLEX RECEPTACLE FLOOR MOUNTED
	SPECIAL PURPOSE RECEPTACLE FLOOR MOUNTED (AMPACITY AS NOTED)
B10	SURFACE MOUNTED RACEWAY LETTER INDICATES TYPE (SEE E600) NUMBER INDICATES LENGTH
J	JUNCTION BOX
	NON FUSED DISCONNECT SWITCH
F	FUSED DISCONNECT SWITCH
	MOTOR STARTER
	COMBINATION MOTOR STARTER
	CONTACTOR
TS	TIME CLOCK SWITCH CONTROL
S_T	TIME SWITCH
S_M	MANUAL MOTOR STARTER SWITCH WITHOUT OVERLOADS
S_{MT}	MANUAL MOTOR STARTER SWITCH WITH THERMAL OVERLOADS
S_{MF}	MANUAL MOTOR STARTER SWITCH WITHOUT OVERLOADS WITH PILOT LIGHT
	PANELBOARD
PB	PUSHBUTTON STATION TYPE AS NOTED
	MUSHROOM SWITCH
	MOTOR (HORSEPOWER INDICATED ON PLANS)
T	TRANSFORMER
	GENERATOR
MOD	MOTOR OPERATED DAMPER
T	THERMOSTAT
UPS	UNINTERRUPTIBLE POWER SUPPLY
SPD	SURGE PROTECTION DEVICE
	UNIT HEATER
VFD	VARIABLE FREQUENCY DRIVE

LINES & REFERENCE SYMBOLS

	NEW WORK LINEWEIGHT
	DEMOLITION WORK LINEWEIGHT
	EXISTING WORK LINEWEIGHT
	FUTURE WORK LINEWEIGHT
	POINT OF CONNECTION
	TERMINATION POINT OF DEMOLITION
	POINT OF FUTURE CONNECTION
#	SHEET NOTE REFERENCE
#	REVISION REFERENCE
E-1	EQUIPMENT TAG
	NORTH ARROW
1 VIEW TITLE SCALE:	INDICATES PLAN/DETAIL NUMBER PLAN/DETAIL TITLE SCALE (AS REQUIRED)
1 M301	INDICATES PLAN/DETAIL NUMBER INDICATES DRAWING ON WHICH DRAWING SECTION APPEARS

GROUNDING

	GROUND ROD
G	BARE COPPER CONDUCTOR EXPOSED
G	BARE COPPER CONDUCTOR BURIED OR EMBEDDED IN CONCRETE
GND	GROUND BUS
TW	TEST WELL

CONDUIT FEEDERS & BRANCH CIRCUITS

	CONDUIT - EXPOSED
	CONDUIT TURNED
	CONDUIT TURNED UP
	CONDUIT TURNED DOWN
	CONDUIT CAPPED
	CIRCUIT HOME RUN
10	CONDUIT FEEDER IDENTIFICATION (SEE FEEDER LEGEND ON DRAWING)
3 #8, 1 #10 EGC - 3/4"C.	CONDUIT FEEDER IDENTIFICATION

GENERAL NOTES

1. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH THE NATIONAL ELECTRICAL CODE (NFPA-70) AS ADOPTED AND AMMENDED BY PHILADELPHIA ELECTRICAL CODE AND THE THE COMMONWEALTH OF PENNSYLVANIA.
2. ALL ELECTRICAL WORK SHALL BE PROPERLY GROUNDED AND SHALL MEET ALL REQUIREMENTS OF THE APPLICABLE SECTION OF THE NATIONAL ELECTRICAL CODE (NEC) AND ANY AUTHORITIES HAVING JURISDICTION.
3. DRAWINGS ARE DIAGRAMMATIC IN NATURE, THE CONTRACTOR SHALL VERIFY DIMENSIONS PRIOR TO INSTALLATION. CONTRACTOR SHALL COORDINATE ALL WORK WITH OTHER DIVISIONS TRADES TO PROVIDE A COMPLETE AND OPERATIONAL SYSTEM. CONTRACTOR SHALL COORDINATE LOCATION OF FIXTURES, DEVICES, ETC WITH OTHER TRADES IN ORDER TO AVOID INTERFERENCES.
4. ARCHITECTURAL FEATURES SHOWN ON THESE DRAWINGS ARE FOR BACKGROUND INFORMATION ONLY. REFER TO THE ARCHITECTURAL AND STRUCTURAL DRAWINGS FOR ACTUAL BUILDING CONSTRUCTION WALLS AND CURBS. REFER TO EQUIPMENT DRAWINGS FOR ACTUAL LOCATION OF EQUIPMENT.
5. EXACT CONDUIT STUB-UP LOCATIONS ARE TO BE DETERMINED BY THE ELECTRICAL CONTRACTOR BASED ON THE CERTIFIED MANUFACTURER'S DRAWINGS OF RESPECTIVE EQUIPMENT. CONDUITS SHALL BE INSTALLED TO AGREE WITH EQUIPMENT FURNISHED.
6. WALL AND FLOOR PENETRATIONS SHALL BE BY THE ELECTRICAL CONTRACTOR. PROVIDE FIRESTOP AS REQUIRED FOR ALL PENETRATIONS MADE FOR ELECTRICAL WORK.
7. COORDINATE THE COLORS OF WIRING DEVICES AND FACEPLATES WITH THE ARCHITECT.
8. EQUIPMENT AND DEVICES SHALL MATCH THE BUILDING STANDARD.

WIRING METHODS

1. INDOORS
   A. IN FINISHED AREAS ALL RACEWAY AND WIRING SHALL BE CONCEALED AND BOXES RECESSES. WIRING INSTALLED IN MASONRY WALLS SHALL BE EMT OR GALVANIZED RIGID STEEL. WIRING INSTALLED IN STUD WALL CAVITIES OR ABOVE HUNG CEILINGS MAY BE TYPE MC CABLE OR EMT.
   B. IN UNFINISHED AREAS SUCH AS MECHANICAL AND ELECTRICAL ROOMS WIRING SHALL BE INSTALLED IN EMT.
   C. WIRING IN THE CRAWLSPACE SHALL BE INSTALLED IN RIGID METAL CONDUIT.

GANNETT FLEMING

Gannett Fleming, Inc.

1010 Adams Avenue
Valley Forge • PA • 19403
t 610 • 650 • 8156

www.gannettfleming.com

READING FIRE DEPARTMENT, ENGINE 7

950 MCKNIGHT ST, READING, PA 19601

HVAC REPLACEMENT / IAQ UPGRADE

NO.	DATE	DESCRIPTION

REVISIONS

BID DOCUMENTS

PROJECT NUMBER: 066446.007

DESIGNED:	MODELED:	CHECKED:
FJR	AMF	BAM

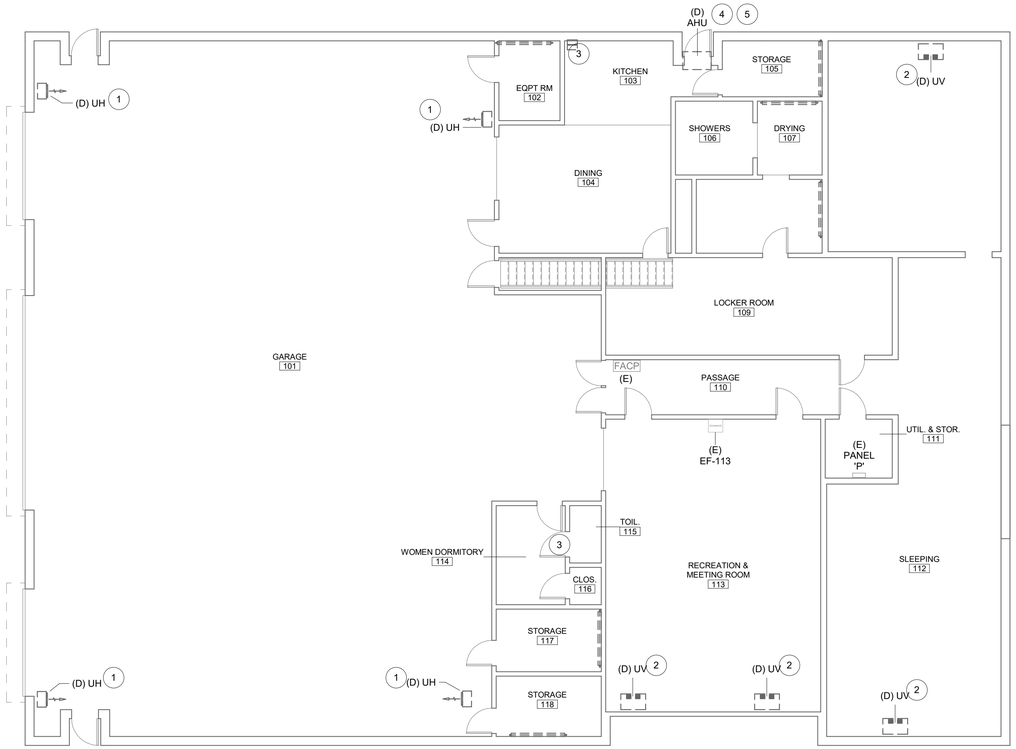
DATE: 22 JULY 2022

DRAWING TITLE: ELECTRICAL GENERAL NOTES, LEGENDS AND ABBREVIATIONS

DRAWING NO.: SHEET 14 OF 17

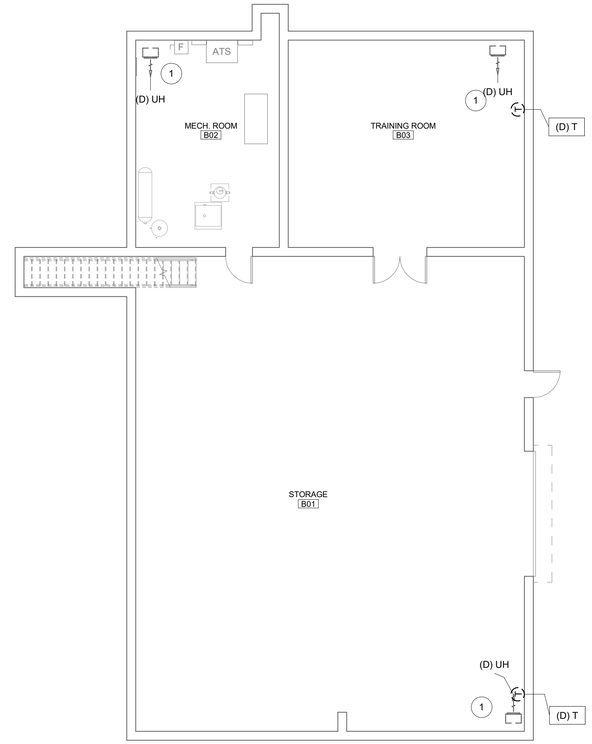
E001

Local File:BIM 360:///066446-City_of_Reading/066446_007_MEP_Central_R21.rvt
Project Number: 066446.007
7/22/2022 7:17:07 PM



1 ELECTRICAL FIRST FLOOR DEMOLITION PLAN

SCALE: 1/8" = 1'-0"



2 ELECTRICAL BASEMENT DEMOLITION PLAN

SCALE: 1/8" = 1'-0"

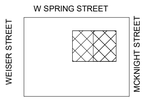
DEMOLITION SHEET KEYNOTES

1. DISCONNECT HYDRONIC UNIT HEATER WIRING AND CONDUIT AND RETAIN FOR REUSE. MAINTAIN EXISTING CIRCUITS FOR CONNECTION OF NEW HYDRONIC UNIT HEATER (UH).
2. DISCONNECT CABINET UNIT VENTILATOR WIRING AND CONDUIT AND RETAIN FOR REUSE. MAINTAIN EXISTING CIRCUITS FOR CONNECTION OF NEW CABINET UNIT VENTILATOR (CUV).
3. DISCONNECT EXHAUST FAN WIRING AND CONDUIT AND RETAIN FOR REUSE. MAINTAIN EXISTING CIRCUITS FOR CONNECTION OF NEW EXHAUST FAN (EF).
4. DISCONNECT AIR HANDLING UNIT WIRING AND CONDUIT AND RETAIN FOR REUSE. MAINTAIN EXISTING CIRCUITS FOR CONNECTION OF NEW AIR HANDLING UNIT (AHU).
5. DISCONNECT AIR-COOLED CONDENSING UNIT WIRING AND CONDUIT AND RETAIN FOR REUSE. MAINTAIN EXISTING CIRCUITS FOR CONNECTION OF NEW AIR-COOLED CONDENSING UNIT (ACCU).

GENERAL DEMOLITION NOTES

1. SEE ELECTRICAL DRAWING E001 FOR GENERAL NOTES AND PROJECT INFORMATION.
2. NOTES AND GRAPHIC REPRESENTATIONS SHALL NOT LIMIT THE EXTENT OR DEMOLITION REQUIRED. CONTRACTOR SHALL VISIT THE SITE, CAREFULLY EXAMINE EXISTING CONDITIONS AND SHALL PERFORM ALL DEMOLITION REQUIRED TO ACHIEVE THE FINAL DESIGN INTENT AS REQUIRED BY THE CONTRACT DOCUMENTS. EXTENT OF ALL DEMOLITION WORK SHALL BE COORDINATED WITH THE OWNER & ENGINEER.
3. ALL WORK REQUIRED TO REMAIN IN SERVICE BUT INTERFERING WITH THE ALTERATIONS SHALL BE RELOCATED AND RECONNECTED USING MATERIALS AND STANDARDS OF THIS CONTRACT.
4. EQUIPMENT AND WIRING TO BE REMOVED SHALL BE DE-ENERGIZED PRIOR TO ANY DEMOLITION WORK.
5. EQUIPMENT INDICATED TO BE REMOVED SHALL BE TAKEN FROM THE SITE AND DISPOSED OF IN ACCORDANCE WITH APPLICABLE LAWS AND ENVIRONMENTAL REGULATIONS. EQUIPMENT REQUIRED TO BE TURNED OVER TO THE OWNER, SHALL BE PLACED IN A MUTUALLY ACCEPTED LOCATION.
6. PROCEDURE FOR FEEDERS AND BRANCH CIRCUITS TO BE REMOVED: CONDUIT AND SUPPORTS SHALL BE REMOVED TO THE PANEL OF ORIGIN OR THE BOUNDARY OF THE PROJECT AREA. WIRING SHALL BE REMOVED TO THE PANEL OF ORIGIN. WHERE EMPTY CONDUITS REMAIN, INSTALL A PULL STRING AND IDENTIFY AT BOTH ENDS.
7. PROCEDURE FOR FEEDERS AND BRANCH CIRCUITS TO BE RE-USED: REMOVE CONDUIT AND WIRING TO LOCATIONS WHICH AVOID CONFLICTS WITH NEW WORK. INSTALL JUNCTION BOXES, TAPE OFF CONDUCTORS AND IDENTIFY WITH PANEL AND CIRCUIT NUMBER.
8. WHERE PORTIONS OF EXISTING BRANCH CIRCUITS ARE REMOVED, MAINTAIN THE CONTINUITY OF CIRCUITING TO REMAINING DEVICES.
9. PROVIDE BLANK COVER PLATES AT OPEN BOXES WHERE EXISTING RECEPTACLES OR ELECTRICAL DEVICES ARE REMOVED FROM ENCLOSURES OR SURFACES NOT INDICATED TO BE REPAIRED OR REFINISHED.
10. PROVIDE ADDITIONAL SUPPORT FOR ALL EXISTING CONDUITS, LOW VOLTAGE CABLING AND DEVICES TO REMAIN WHICH ARE AFFECTED BY DEMOLITION OF EXISTING CEILINGS AND PARTITIONS.
11. ALL WORK SHALL BE PROPERLY IDENTIFIED AFTER DEMOLITION. UPDATE ALL PANEL SCHEDULES TO REFLECT EQUIPMENT AND CIRCUIT REMOVALS.

0' 8' 16'



KEYPLAN

GANNETT FLEMING

Gannett Fleming, Inc.

1010 Adams Avenue
Valley Forge • PA • 19403
t 610 • 650 • 8156

www.gannettfleming.com

READING FIRE DEPARTMENT, ENGINE 7

950 MCKNIGHT ST, READING, PA 19601

HVAC REPLACEMENT / IAQ UPGRADE

NO.	DATE	DESCRIPTION

REVISIONS

BID DOCUMENTS

PROJECT NUMBER: 066446.007

DESIGNED:	MODELED:	CHECKED:
FJR	AMF	BAM

DATE: 22 JULY 2022

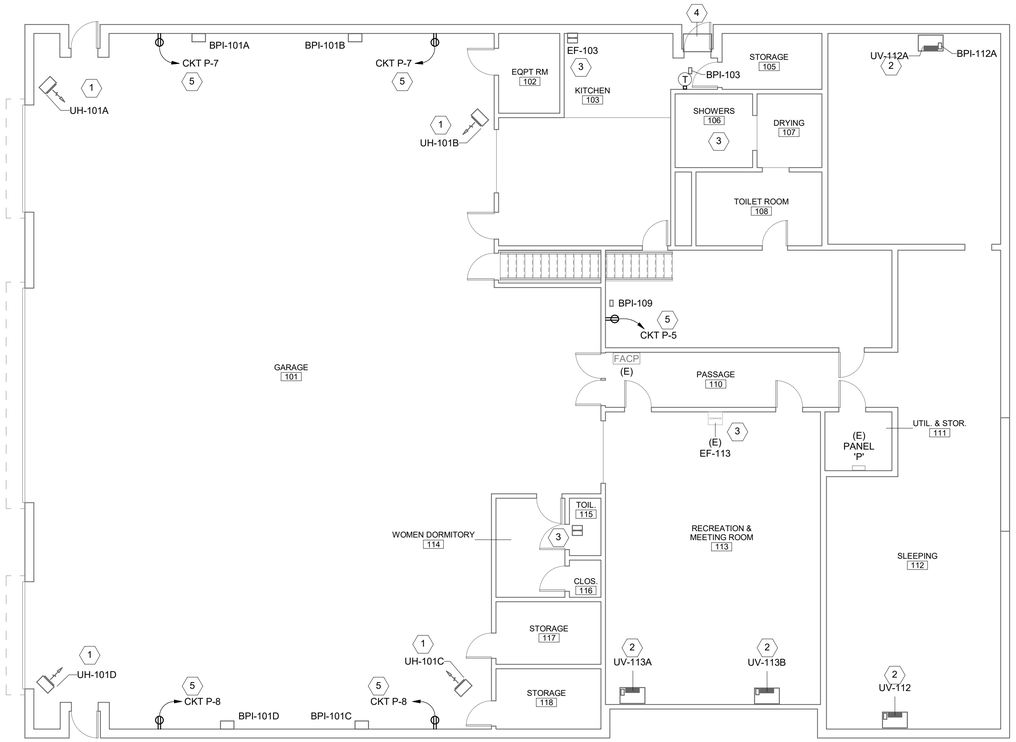
DRAWING TITLE: ELECTRICAL DEMOLITION BASEMENT AND FIRST FLOOR

DRAWING NO.: SHEET 15 OF 17

E101

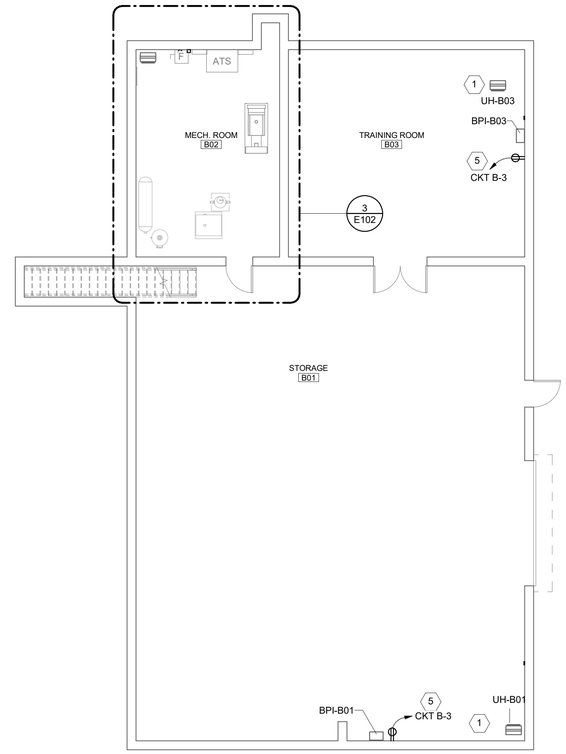
Local File:BIM 360:///066446-City_of_Reading/066446_007_MEP_Central_R21.rvt

Project Number: 066446.007
7/22/2022 7:17:08 PM



1 ELECTRICAL FIRST FLOOR NEW WORK PLAN

SCALE: 1/8" = 1'-0"



2 ELECTRICAL BASEMENT NEW WORK PLAN

SCALE: 1/8" = 1'-0"

NEW WORK GENERAL NOTES

1. REFER TO DRAWING E001 FOR GENERAL NOTES AND WIRING METHODS.
2. ALL ELECTRICAL EQUIPMENT SHOWN ON DRAWING IS NEW UNLESS NOTED OTHERWISE.

NEW WORK SHEET KEYNOTES

1. RECONNECT HYDRONIC UNIT HEATER WIRING AND CONDUIT TO NEW HYDRONIC UNIT HEATER (UH). EXTEND WIRING/CONDUIT AS NEEDED.
2. RECONNECT CABINET UNIT VENTILATOR WIRING AND CONDUIT TO NEW CABINET UNIT VENTILATOR (CUV). EXTEND WIRING/CONDUIT AS NEEDED.
3. RECONNECT EXHAUST FAN WIRING AND CONDUIT REUSE. MAINTAIN TO NEW EXHAUST FAN (EF). EXTEND WIRING/CONDUIT AS NEEDED.
4. NEW AIR HANDLING UNIT (AHU) OR FAN COIL UNIT (FCUI). CONNECT TO CIRCUIT SERVING AHU-211. EXTEND WIRING/CONDUIT AS NEEDED.
5. PROVIDE 120V DUPLEX RECEPTACLE FOR NEW BPI. CONNECT TO A 20/1 CIRCUIT BREAKER IN PANEL P. PROVIDE (2) #12, (1) #12GRD IN 3/4 CONDUIT.

PANEL DESIGNATION:

(E) PANEL 'P'

LOCATION: UTIL. & STOR. 111
FED FROM:

TYPE: BRANCH...
NUMBER OF POLES: 42
MAIN BUS RATINGS: 225.0
MAINS RATING: 225A

VOLTAGE: 120/240V, 1PH,...
PANEL MOUNTING: RECESSED
PANEL ENCLOSURE:
PANEL MIN. A.I.C. RATING:

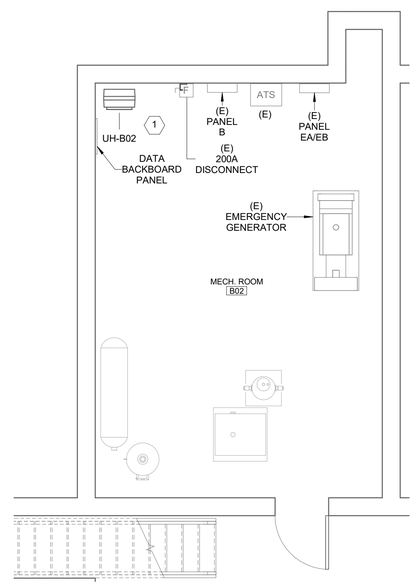
CIR. No.	CIR. BKR.	DESCRIPTION	COND.	GRND.	WIRE SIZE	LOAD KVA AØ	LOAD KVA BØ	LOAD KVA AØ	LOAD KVA BØ	WIRE SIZE	GRND.	COND.	DESCRIPTION	CIR. BKR.	CIR. No.
1	20	KITCHEN COUNTER RECEPTACLES	--	--	--	0.00		0.00		--	--	--	ENGINE RM RECEPTS & UNIT HEATERS	20	2
3	20	REFRIGERATOR	--	--	--		0.00		0.00	--	--	--	LOCKER ROOM SHOWER ROOM...	20	4
5	20	FIRE PANEL RECEPTACLE	--	--	--	0.00		0.00		--	--	--	DINING ROOM LIGHTS	20	6
7	20	**BPI-101A & B RECEPTACLES - ROOM...**	--	--	--		1.80		1.80	--	--	--	**BPI-101C & D RECEPTACLES - ROOM...**	20	8
9	20	MEETING ROOM LIGHTS	--	--	--	0.00		0.00		--	--	--	MEETING RM RECEPTS & WATER...	20	10
11	20	CHIEF TOILET FAN & RECEPTACLES	--	--	--		0.00		0.00	--	--	--	MEETING ROOM RECEPTACLES	20	12
13	20	LOCKER ROOM RECEPTACLES & EF	--	--	--	0.00		0.00		--	--	--	MEET WXH FAN & LOCKER...	20	14
15	20	SLEEPING AREA RECEPTACLES	--	--	--		0.00		0.00	--	--	--	SLEEPING AREA RECEPTACLES	20	16
17	20	SLEEPING AREA RECEPTACLES	--	--	--	0.00		0.00		--	--	--	SLEEPING AREA LIGHTS	20	18
19	20	SLEEPING AREA LIGHTS	--	--	--		0.00		0.00	--	--	--	ENGINE ROOM LIGHTS	20	20
21	20	ENGINE ROOM LIGHTS	--	--	--	0.00		0.00		--	--	--	SE ENGINE RM HEATER &...	20	22
23	20	FIRE PANEL	--	--	--		0.00		0.00	--	--	--	OFFICE RECEPTACLES & LIGHTS	20	24
25	20	VENDING MACHINE	--	--	--	0.00		0.00		--	--	--	EXTRACTOR	20	26
27	30	HEATER	--	--	--		0.00		0.00	--	--	--	2 POLE ZONE CONTROL	15	28
29						0.00		0.00							30
31	20	AIR COND. MEETING ROOM	--	--	--		0.00		0.00	--	--	--	A/C KITCHEN	30	32
33						0.00		0.00							34
35	20	AIR COND. SLEEPING QUATERS	--	--	--		0.00		0.00	--	--	--	AIR COND. MEETING ROOM	20	36
37						0.00		0.00							38
39	20	AIR COND. SLEEPING QUATERS	--	--	--		0.00		0.00	--	--	--	RANGE	50	40
41						0.00		0.00							42

CONNECTED LOAD - KVA

AØ	0.00
BØ	3.60
CØ	0.00
	3.60 *KVA TOTAL*
	3.60 **TOTAL ESTIMATED DEMAND**

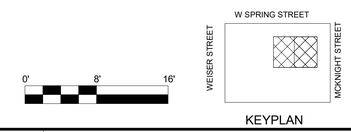
X **SOLID NEUTRAL BUS**
X **EQUIPMENT GROUND BUS**
INTERNAL/EXTERNAL SURGE PROTECTION...
DIGITAL POWER METER
FEED THROUGH LUGS
SHUNT TRIP

NOTES:
1.
2.
3.
4.
5.



3 ELECTRICAL ENLARGED ELECTRICAL ROOM

SCALE: 1/4" = 1'-0"



KEYPLAN

GANNETT FLEMING

Gannett Fleming, Inc.

1010 Adams Avenue
Valley Forge • PA • 19403
t 610 • 650 • 8156

www.**gannettfleming**.com

READING FIRE DEPARTMENT, ENGINE 7

950 MCKNIGHT ST, READING, PA 19601

HVAC REPLACEMENT / IAQ UPGRADE

NO.	DATE	DESCRIPTION

REVISIONS

BID DOCUMENTS

PROJECT NUMBER: 066446.007

DESIGNED:	MODELED:	CHECKED:
FJR	AMF	BAM

DATE: 22 JULY 2022

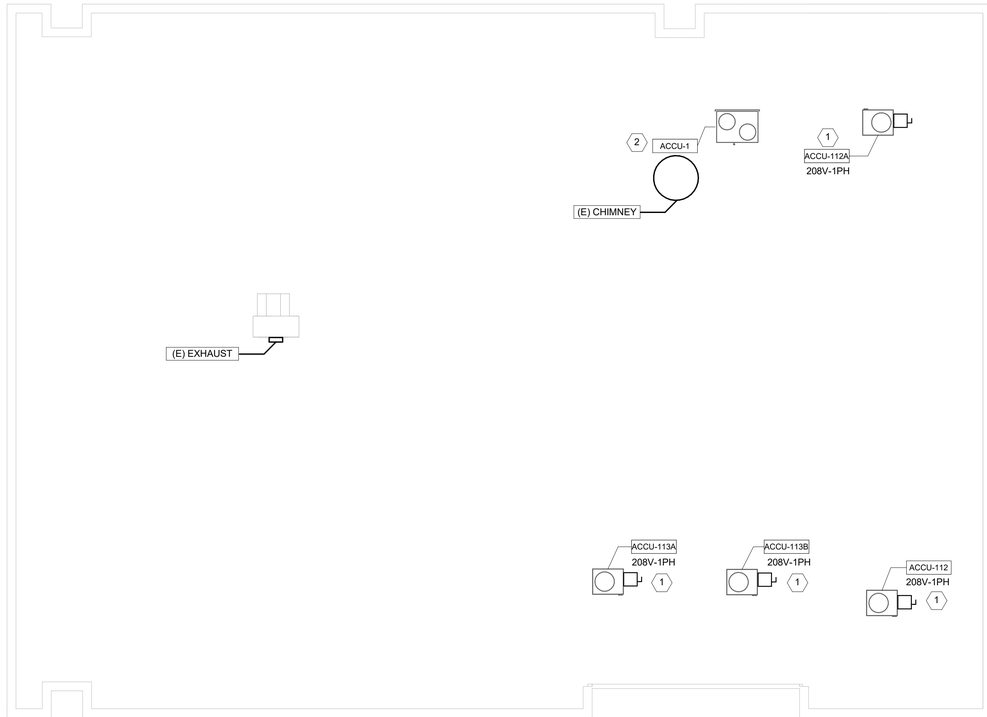
DRAWING TITLE: ELECTRICAL

NEW WORK BASEMENT AND FIRST FLOOR

DRAWING NO.: SHEET 16 OF 17

E102

Local File:BIM 360://066446-City_of_Reading/066446_007_MEP_Central_R21.rvt
Project Number: 066446.007
7/22/2022 7:17:09 PM



1 ELECTRICAL ROOF PLAN

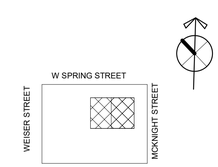
SCALE: 1/8" = 1'-0"

NEW WORK GENERAL NOTES

1. REFER TO DRAWING E001 FOR GENERAL NOTES AND WIRING METHODS.
2. ALL ELECTRICAL EQUIPMENT SHOWN ON DRAWING IS NEW UNLESS NOTED OTHERWISE.

NEW WORK SHEET KEYNOTES

1. PROVIDE NEW DEDICATED CIRCUIT (CONDUIT AND WIRE) TO PANEL EA .
2. RECONNECT CONDENSING UNIT WIRING AND CONDUIT REUSE. EXTEND WIRING/CONDUIT AS NEEDED.



KEYPLAN

GANNETT FLEMING

Gannett Fleming, Inc.

1010 Adams Avenue
Valley Forge • PA • 19403
t 610 • 650 • 8156

www.gannettfleming.com

READING FIRE DEPARTMENT, ENGINE 7

950 MCKNIGHT ST, READING, PA 19601

HVAC REPLACEMENT / IAQ UPGRADE

NO.	DATE	DESCRIPTION

REVISIONS

BID DOCUMENTS

PROJECT NUMBER: 066446.007

DESIGNED:	MODELED:	CHECKED:
FJR	AMF	MJM

DATE: 22 JULY 2022

DRAWING TITLE: ELECTRICAL ROOF PLAN

DRAWING NO.: E103

SHEET 17 OF 17

Local File:BIM 360://066446-City_of_Reading/066446_007_MEP_Central_R21.rvt

Project Number: 066446.007
7/22/2022 7:17:10 PM